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The Transport Layer Security (TLS) Protocol Version 1.2

Status of This Memo

This document specifies an Internet standards track protocol for the Internet community, and requests discussion and suggestions for improvements. Please refer to the current edition of the "Internet Official Protocol Standards" (STD 1) for the standardization state and status of this protocol. Distribution of this memo is unlimited.

Abstract

This document specifies Version 1.2 of the Transport Layer Security (TLS) protocol. The TLS protocol provides communications security over the Internet. The protocol allows client/server applications to communicate in a way that is designed to prevent eavesdropping, tampering, or message forgery.

Table of Contents

1. Introduction	4
1.1. Requirements Terminology	5
1.2. Major Differences from TLS 1.1	5
2. Goals	6
3. Goals of This Document	7
4. Presentation Language	7
4.1. Basic Block Size	7
4.2. Miscellaneous	8
4.3. Vectors	8
4.4. Numbers	9
4.5. Enumerateds	9
4.6. Constructed Types	10
4.6.1. Variants	10
4.7. Cryptographic Attributes	12
4.8. Constants	14
5. HMAC and the Pseudorandom Function	14
6. The TLS Record Protocol	15
6.1. Connection States	16
6.2. Record Layer	19
6.2.1. Fragmentation	19

6.2.2.	Record Compression and Decompression	20
6.2.3.	Record Payload Protection	21
6.2.3.1.	Null or Standard Stream Cipher	22
6.2.3.2.	CBC Block Cipher	22
6.2.3.3.	AEAD Ciphers	24
6.3.	Key Calculation	25
7.	The TLS Handshaking Protocols	26
7.1.	Change Cipher Spec Protocol	27
7.2.	Alert Protocol	28
7.2.1.	Closure Alerts	29
7.2.2.	Error Alerts	30
7.3.	Handshake Protocol Overview	33
7.4.	Handshake Protocol	37
7.4.1.	Hello Messages	38
7.4.1.1.	Hello Request	38
7.4.1.2.	Client Hello	39
7.4.1.3.	Server Hello	42
7.4.1.4.	Hello Extensions	44
7.4.1.4.1.	Signature Algorithms	45
7.4.2.	Server Certificate	47
7.4.3.	Server Key Exchange Message	50
7.4.4.	Certificate Request	53
7.4.5.	Server Hello Done	55
7.4.6.	Client Certificate	55
7.4.7.	Client Key Exchange Message	57
7.4.7.1.	RSA-Encrypted Premaster Secret Message	58
7.4.7.2.	Client Diffie-Hellman Public Value	61
7.4.8.	Certificate Verify	62
7.4.9.	Finished	63
8.	Cryptographic Computations	64
8.1.	Computing the Master Secret	64
8.1.1.	RSA	65
8.1.2.	Diffie-Hellman	65
9.	Mandatory Cipher Suites	65
10.	Application Data Protocol	65
11.	Security Considerations	65
12.	IANA Considerations	65
Appendix A.	Protocol Data Structures and Constant Values	68
A.1.	Record Layer	68
A.2.	Change Cipher Specs Message	69
A.3.	Alert Messages	69
A.4.	Handshake Protocol	70
A.4.1.	Hello Messages	71
A.4.2.	Server Authentication and Key Exchange Messages	72
A.4.3.	Client Authentication and Key Exchange Messages	74
A.4.4.	Handshake Finalization Message	74
A.5.	The Cipher Suite	75
A.6.	The Security Parameters	77

A.7. Changes to RFC 4492	78
Appendix B. Glossary	78
Appendix C. Cipher Suite Definitions	83
Appendix D. Implementation Notes	85
D.1. Random Number Generation and Seeding	85
D.2. Certificates and Authentication	85
D.3. Cipher Suites	85
D.4. Implementation Pitfalls	85
Appendix E. Backward Compatibility	87
E.1. Compatibility with TLS 1.0/1.1 and SSL 3.0	87
E.2. Compatibility with SSL 2.0	88
E.3. Avoiding Man-in-the-Middle Version Rollback	90
Appendix F. Security Analysis	91
F.1. Handshake Protocol	91
F.1.1. Authentication and Key Exchange	91
F.1.1.1. Anonymous Key Exchange	91
F.1.1.2. RSA Key Exchange and Authentication	92
F.1.1.3. Diffie-Hellman Key Exchange with Authentication	92
F.1.2. Version Rollback Attacks	93
F.1.3. Detecting Attacks Against the Handshake Protocol ..	94
F.1.4. Resuming Sessions	94
F.2. Protecting Application Data	94
F.3. Explicit IVs	95
F.4. Security of Composite Cipher Modes	95
F.5. Denial of Service	96
F.6. Final Notes	96
Normative References	97
Informative References	98
Working Group Information	101
Contributors	101

1. Introduction

The primary goal of the TLS protocol is to provide privacy and data integrity between two communicating applications. The protocol is composed of two layers: the TLS Record Protocol and the TLS Handshake Protocol. At the lowest level, layered on top of some reliable transport protocol (e.g., TCP [TCP]), is the TLS Record Protocol. The TLS Record Protocol provides connection security that has two basic properties:

- The connection is private. Symmetric cryptography is used for data encryption (e.g., AES [AES], RC4 [SCH], etc.). The keys for this symmetric encryption are generated uniquely for each connection and are based on a secret negotiated by another protocol (such as the TLS Handshake Protocol). The Record Protocol can also be used without encryption.
- The connection is reliable. Message transport includes a message integrity check using a keyed MAC. Secure hash functions (e.g., SHA-1, etc.) are used for MAC computations. The Record Protocol can operate without a MAC, but is generally only used in this mode while another protocol is using the Record Protocol as a transport for negotiating security parameters.

The TLS Record Protocol is used for encapsulation of various higher-level protocols. One such encapsulated protocol, the TLS Handshake Protocol, allows the server and client to authenticate each other and to negotiate an encryption algorithm and cryptographic keys before the application protocol transmits or receives its first byte of data. The TLS Handshake Protocol provides connection security that has three basic properties:

- The peer's identity can be authenticated using asymmetric, or public key, cryptography (e.g., RSA [RSA], DSA [DSS], etc.). This authentication can be made optional, but is generally required for at least one of the peers.
- The negotiation of a shared secret is secure: the negotiated secret is unavailable to eavesdroppers, and for any authenticated connection the secret cannot be obtained, even by an attacker who can place himself in the middle of the connection.
- The negotiation is reliable: no attacker can modify the negotiation communication without being detected by the parties to the communication.

One advantage of TLS is that it is application protocol independent. Higher-level protocols can layer on top of the TLS protocol transparently. The TLS standard, however, does not specify how protocols add security with TLS; the decisions on how to initiate TLS handshaking and how to interpret the authentication certificates exchanged are left to the judgment of the designers and implementors of protocols that run on top of TLS.

1.1. Requirements Terminology

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC 2119 [REQ].

1.2. Major Differences from TLS 1.1

This document is a revision of the TLS 1.1 [TLS1.1] protocol which contains improved flexibility, particularly for negotiation of cryptographic algorithms. The major changes are:

- The MD5/SHA-1 combination in the pseudorandom function (PRF) has been replaced with cipher-suite-specified PRFs. All cipher suites in this document use P_SHA256.
- The MD5/SHA-1 combination in the digitally-signed element has been replaced with a single hash. Signed elements now include a field that explicitly specifies the hash algorithm used.
- Substantial cleanup to the client's and server's ability to specify which hash and signature algorithms they will accept. Note that this also relaxes some of the constraints on signature and hash algorithms from previous versions of TLS.
- Addition of support for authenticated encryption with additional data modes.
- TLS Extensions definition and AES Cipher Suites were merged in from external [TLSEXT] and [TLSAES].
- Tighter checking of EncryptedPreMasterSecret version numbers.
- Tightened up a number of requirements.
- Verify_data length now depends on the cipher suite (default is still 12).
- Cleaned up description of Bleichenbacher/Klima attack defenses.

- Alerts MUST now be sent in many cases.
- After a `certificate_request`, if no certificates are available, clients now MUST send an empty certificate list.
- `TLS_RSA_WITH_AES_128_CBC_SHA` is now the mandatory to implement cipher suite.
- Added HMAC-SHA256 cipher suites.
- Removed IDEA and DES cipher suites. They are now deprecated and will be documented in a separate document.
- Support for the SSLv2 backward-compatible hello is now a MAY, not a SHOULD, with sending it a SHOULD NOT. Support will probably become a SHOULD NOT in the future.
- Added limited "fall-through" to the presentation language to allow multiple case arms to have the same encoding.
- Added an Implementation Pitfalls sections
- The usual clarifications and editorial work.

2. Goals

The goals of the TLS protocol, in order of priority, are as follows:

1. Cryptographic security: TLS should be used to establish a secure connection between two parties.
2. Interoperability: Independent programmers should be able to develop applications utilizing TLS that can successfully exchange cryptographic parameters without knowledge of one another's code.
3. Extensibility: TLS seeks to provide a framework into which new public key and bulk encryption methods can be incorporated as necessary. This will also accomplish two sub-goals: preventing the need to create a new protocol (and risking the introduction of possible new weaknesses) and avoiding the need to implement an entire new security library.
4. Relative efficiency: Cryptographic operations tend to be highly CPU intensive, particularly public key operations. For this reason, the TLS protocol has incorporated an optional session caching scheme to reduce the number of connections that need to be established from scratch. Additionally, care has been taken to reduce network activity.

3. Goals of This Document

This document and the TLS protocol itself are based on the SSL 3.0 Protocol Specification as published by Netscape. The differences between this protocol and SSL 3.0 are not dramatic, but they are significant enough that the various versions of TLS and SSL 3.0 do not interoperate (although each protocol incorporates a mechanism by which an implementation can back down to prior versions). This document is intended primarily for readers who will be implementing the protocol and for those doing cryptographic analysis of it. The specification has been written with this in mind, and it is intended to reflect the needs of those two groups. For that reason, many of the algorithm-dependent data structures and rules are included in the body of the text (as opposed to in an appendix), providing easier access to them.

This document is not intended to supply any details of service definition or of interface definition, although it does cover select areas of policy as they are required for the maintenance of solid security.

4. Presentation Language

This document deals with the formatting of data in an external representation. The following very basic and somewhat casually defined presentation syntax will be used. The syntax draws from several sources in its structure. Although it resembles the programming language "C" in its syntax and XDR [XDR] in both its syntax and intent, it would be risky to draw too many parallels. The purpose of this presentation language is to document TLS only; it has no general application beyond that particular goal.

4.1. Basic Block Size

The representation of all data items is explicitly specified. The basic data block size is one byte (i.e., 8 bits). Multiple byte data items are concatenations of bytes, from left to right, from top to bottom. From the byte stream, a multi-byte item (a numeric in the example) is formed (using C notation) by:

```
value = (byte[0] << 8*(n-1)) | (byte[1] << 8*(n-2)) |  
        ... | byte[n-1];
```

This byte ordering for multi-byte values is the commonplace network byte order or big-endian format.

4.2. Miscellaneous

Comments begin with `/*` and end with `*/`.

Optional components are denoted by enclosing them in `"[[ ]]"` double brackets.

Single-byte entities containing uninterpreted data are of type `opaque`.

4.3. Vectors

A vector (single-dimensioned array) is a stream of homogeneous data elements. The size of the vector may be specified at documentation time or left unspecified until runtime. In either case, the length declares the number of bytes, not the number of elements, in the vector. The syntax for specifying a new type, `T'`, that is a fixed-length vector of type `T` is

```
T T'[n];
```

Here, `T'` occupies `n` bytes in the data stream, where `n` is a multiple of the size of `T`. The length of the vector is not included in the encoded stream.

In the following example, `Datum` is defined to be three consecutive bytes that the protocol does not interpret, while `Data` is three consecutive `Datum`, consuming a total of nine bytes.

```
opaque Datum[3];      /* three uninterpreted bytes */
Datum Data[9];         /* 3 consecutive 3 byte vectors */
```

Variable-length vectors are defined by specifying a subrange of legal lengths, inclusively, using the notation `<floor..ceiling>`. When these are encoded, the actual length precedes the vector's contents in the byte stream. The length will be in the form of a number consuming as many bytes as required to hold the vector's specified maximum (ceiling) length. A variable-length vector with an actual length field of zero is referred to as an empty vector.

```
T T'<floor..ceiling>;
```

In the following example, `mandatory` is a vector that must contain between 300 and 400 bytes of type `opaque`. It can never be empty. The actual length field consumes two bytes, a `uint16`, which is sufficient to represent the value 400 (see Section 4.4). On the other hand, `longer` can represent up to 800 bytes of data, or 400 `uint16` elements, and it may be empty. Its encoding will include a

two-byte actual length field prepended to the vector. The length of an encoded vector must be an even multiple of the length of a single element (for example, a 17-byte vector of uint16 would be illegal).

```
opaque mandatory<300..400>;
/* length field is 2 bytes, cannot be empty */
uint16 longer<0..800>;
/* zero to 400 16-bit unsigned integers */
```

4.4. Numbers

The basic numeric data type is an unsigned byte (uint8). All larger numeric data types are formed from fixed-length series of bytes concatenated as described in Section 4.1 and are also unsigned. The following numeric types are predefined.

```
uint8 uint16[2];
uint8 uint24[3];
uint8 uint32[4];
uint8 uint64[8];
```

All values, here and elsewhere in the specification, are stored in network byte (big-endian) order; the uint32 represented by the hex bytes 01 02 03 04 is equivalent to the decimal value 16909060.

Note that in some cases (e.g., DH parameters) it is necessary to represent integers as opaque vectors. In such cases, they are represented as unsigned integers (i.e., leading zero octets are not required even if the most significant bit is set).

4.5. Enumerateds

An additional sparse data type is available called enum. A field of type enum can only assume the values declared in the definition. Each definition is a different type. Only enumerateds of the same type may be assigned or compared. Every element of an enumerated must be assigned a value, as demonstrated in the following example. Since the elements of the enumerated are not ordered, they can be assigned any unique value, in any order.

```
enum { e1(v1), e2(v2), ... , en(vn) [[, (n)]] } Te;
```

An enumerated occupies as much space in the byte stream as would its maximal defined ordinal value. The following definition would cause one byte to be used to carry fields of type Color.

```
enum { red(3), blue(5), white(7) } Color;
```

One may optionally specify a value without its associated tag to force the width definition without defining a superfluous element.

In the following example, Taste will consume two bytes in the data stream but can only assume the values 1, 2, or 4.

```
enum { sweet(1), sour(2), bitter(4), (32000) } Taste;
```

The names of the elements of an enumeration are scoped within the defined type. In the first example, a fully qualified reference to the second element of the enumeration would be Color.blue. Such qualification is not required if the target of the assignment is well specified.

```
Color color = Color.blue;    /* overspecified, legal */  
Color color = blue;         /* correct, type implicit */
```

For enumerations that are never converted to external representation, the numerical information may be omitted.

```
enum { low, medium, high } Amount;
```

4.6. Constructed Types

Structure types may be constructed from primitive types for convenience. Each specification declares a new, unique type. The syntax for definition is much like that of C.

```
struct {  
    T1 f1;  
    T2 f2;  
    ...  
    Tn fn;  
} [[T]];
```

The fields within a structure may be qualified using the type's name, with a syntax much like that available for enumerations. For example, T.f2 refers to the second field of the previous declaration. Structure definitions may be embedded.

4.6.1. Variants

Defined structures may have variants based on some knowledge that is available within the environment. The selector must be an enumeration type that defines the possible variants the structure defines. There must be a case arm for every element of the enumeration declared in the select. Case arms have limited fall-through: if two case arms follow in immediate succession with no fields in between, then they

both contain the same fields. Thus, in the example below, "orange" and "banana" both contain V2. Note that this is a new piece of syntax in TLS 1.2.

The body of the variant structure may be given a label for reference. The mechanism by which the variant is selected at runtime is not prescribed by the presentation language.

```
struct {
    T1 f1;
    T2 f2;
    ....
    Tn fn;
    select (E) {
        case e1: Te1;
        case e2: Te2;
        case e3: case e4: Te3;
        ....
        case en: Ten;
    } [[fv]];
} [[Tv]];
```

For example:

```
enum { apple, orange, banana } VariantTag;

struct {
    uint16 number;
    opaque string<0..10>; /* variable length */
} V1;

struct {
    uint32 number;
    opaque string[10];    /* fixed length */
} V2;

struct {
    select (VariantTag) { /* value of selector is implicit */
        case apple:
            V1; /* VariantBody, tag = apple */
        case orange:
        case banana:
            V2; /* VariantBody, tag = orange or banana */
    } variant_body; /* optional label on variant */
} VariantRecord;
```

4.7. Cryptographic Attributes

The five cryptographic operations -- digital signing, stream cipher encryption, block cipher encryption, authenticated encryption with additional data (AEAD) encryption, and public key encryption -- are designated digitally-signed, stream-ciphered, block-ciphered, aead-ciphered, and public-key-encrypted, respectively. A field's cryptographic processing is specified by prepending an appropriate key word designation before the field's type specification. Cryptographic keys are implied by the current session state (see Section 6.1).

A digitally-signed element is encoded as a struct DigitallySigned:

```
struct {  
    SignatureAndHashAlgorithm algorithm;  
    opaque signature<0..2^16-1>;  
} DigitallySigned;
```

The algorithm field specifies the algorithm used (see Section 7.4.1.4.1 for the definition of this field). Note that the introduction of the algorithm field is a change from previous versions. The signature is a digital signature using those algorithms over the contents of the element. The contents themselves do not appear on the wire but are simply calculated. The length of the signature is specified by the signing algorithm and key.

In RSA signing, the opaque vector contains the signature generated using the RSASSA-PKCS1-v1_5 signature scheme defined in [PKCS1]. As discussed in [PKCS1], the DigestInfo MUST be DER-encoded [X680] [X690]. For hash algorithms without parameters (which includes SHA-1), the DigestInfo.AlgorithmIdentifier.parameters field MUST be NULL, but implementations MUST accept both without parameters and with NULL parameters. Note that earlier versions of TLS used a different RSA signature scheme that did not include a DigestInfo encoding.

In DSA, the 20 bytes of the SHA-1 hash are run directly through the Digital Signing Algorithm with no additional hashing. This produces two values, r and s. The DSA signature is an opaque vector, as above, the contents of which are the DER encoding of:

```
Dss-Sig-Value ::= SEQUENCE {  
    r INTEGER,  
    s INTEGER  
}
```

Note: In current terminology, DSA refers to the Digital Signature Algorithm and DSS refers to the NIST standard. In the original SSL and TLS specs, "DSS" was used universally. This document uses "DSA" to refer to the algorithm, "DSS" to refer to the standard, and it uses "DSS" in the code point definitions for historical continuity.

In stream cipher encryption, the plaintext is exclusive-ORed with an identical amount of output generated from a cryptographically secure keyed pseudorandom number generator.

In block cipher encryption, every block of plaintext encrypts to a block of ciphertext. All block cipher encryption is done in CBC (Cipher Block Chaining) mode, and all items that are block-ciphered will be an exact multiple of the cipher block length.

In AEAD encryption, the plaintext is simultaneously encrypted and integrity protected. The input may be of any length, and aead-ciphered output is generally larger than the input in order to accommodate the integrity check value.

In public key encryption, a public key algorithm is used to encrypt data in such a way that it can be decrypted only with the matching private key. A public-key-encrypted element is encoded as an opaque vector $\langle 0..2^{16}-1 \rangle$, where the length is specified by the encryption algorithm and key.

RSA encryption is done using the RSAES-PKCS1-v1_5 encryption scheme defined in [PKCS1].

In the following example

```
stream-ciphered struct {
    uint8 field1;
    uint8 field2;
    digitally-signed opaque {
        uint8 field3<0..255>;
        uint8 field4;
    };
} UserType;
```

The contents of the inner struct (field3 and field4) are used as input for the signature/hash algorithm, and then the entire structure is encrypted with a stream cipher. The length of this structure, in bytes, would be equal to two bytes for field1 and field2, plus two bytes for the signature and hash algorithm, plus two bytes for the length of the signature, plus the length of the output of the signing

algorithm. The length of the signature is known because the algorithm and key used for the signing are known prior to encoding or decoding this structure.

4.8. Constants

Typed constants can be defined for purposes of specification by declaring a symbol of the desired type and assigning values to it.

Under-specified types (opaque, variable-length vectors, and structures that contain opaque) cannot be assigned values. No fields of a multi-element structure or vector may be elided.

For example:

```
struct {  
    uint8 f1;  
    uint8 f2;  
} Example1;
```

```
Example1 ex1 = {1, 4}; /* assigns f1 = 1, f2 = 4 */
```

5. HMAC and the Pseudorandom Function

The TLS record layer uses a keyed Message Authentication Code (MAC) to protect message integrity. The cipher suites defined in this document use a construction known as HMAC, described in [HMAC], which is based on a hash function. Other cipher suites MAY define their own MAC constructions, if needed.

In addition, a construction is required to do expansion of secrets into blocks of data for the purposes of key generation or validation. This pseudorandom function (PRF) takes as input a secret, a seed, and an identifying label and produces an output of arbitrary length.

In this section, we define one PRF, based on HMAC. This PRF with the SHA-256 hash function is used for all cipher suites defined in this document and in TLS documents published prior to this document when TLS 1.2 is negotiated. New cipher suites MUST explicitly specify a PRF and, in general, SHOULD use the TLS PRF with SHA-256 or a stronger standard hash function.

First, we define a data expansion function, P_hash(secret, data), that uses a single hash function to expand a secret and seed into an arbitrary quantity of output:

$$\begin{aligned} P_hash(secret, seed) = & HMAC_hash(secret, A(1) + seed) + \\ & HMAC_hash(secret, A(2) + seed) + \\ & HMAC_hash(secret, A(3) + seed) + \dots \end{aligned}$$

where + indicates concatenation.

A() is defined as:

$$\begin{aligned} A(0) &= seed \\ A(i) &= HMAC_hash(secret, A(i-1)) \end{aligned}$$

P_hash can be iterated as many times as necessary to produce the required quantity of data. For example, if P_SHA256 is being used to create 80 bytes of data, it will have to be iterated three times (through A(3)), creating 96 bytes of output data; the last 16 bytes of the final iteration will then be discarded, leaving 80 bytes of output data.

TLS's PRF is created by applying P_hash to the secret as:

$$PRF(secret, label, seed) = P_hash(secret, label + seed)$$

The label is an ASCII string. It should be included in the exact form it is given without a length byte or trailing null character. For example, the label "slithy toves" would be processed by hashing the following bytes:

73 6C 69 74 68 79 20 74 6F 76 65 73

6. The TLS Record Protocol

The TLS Record Protocol is a layered protocol. At each layer, messages may include fields for length, description, and content. The Record Protocol takes messages to be transmitted, fragments the data into manageable blocks, optionally compresses the data, applies a MAC, encrypts, and transmits the result. Received data is decrypted, verified, decompressed, reassembled, and then delivered to higher-level clients.

Four protocols that use the record protocol are described in this document: the handshake protocol, the alert protocol, the change cipher spec protocol, and the application data protocol. In order to allow extension of the TLS protocol, additional record content types can be supported by the record protocol. New record content type values are assigned by IANA in the TLS Content Type Registry as described in Section 12.

Implementations MUST NOT send record types not defined in this document unless negotiated by some extension. If a TLS implementation receives an unexpected record type, it MUST send an `unexpected_message` alert.

Any protocol designed for use over TLS must be carefully designed to deal with all possible attacks against it. As a practical matter, this means that the protocol designer must be aware of what security properties TLS does and does not provide and cannot safely rely on the latter.

Note in particular that type and length of a record are not protected by encryption. If this information is itself sensitive, application designers may wish to take steps (padding, cover traffic) to minimize information leakage.

6.1. Connection States

A TLS connection state is the operating environment of the TLS Record Protocol. It specifies a compression algorithm, an encryption algorithm, and a MAC algorithm. In addition, the parameters for these algorithms are known: the MAC key and the bulk encryption keys for the connection in both the read and the write directions. Logically, there are always four connection states outstanding: the current read and write states, and the pending read and write states. All records are processed under the current read and write states. The security parameters for the pending states can be set by the TLS Handshake Protocol, and the `ChangeCipherSpec` can selectively make either of the pending states current, in which case the appropriate current state is disposed of and replaced with the pending state; the pending state is then reinitialized to an empty state. It is illegal to make a state that has not been initialized with security parameters a current state. The initial current state always specifies that no encryption, compression, or MAC will be used.

The security parameters for a TLS Connection read and write state are set by providing the following values:

connection end

Whether this entity is considered the "client" or the "server" in this connection.

PRF algorithm

An algorithm used to generate keys from the master secret (see Sections 5 and 6.3).

bulk encryption algorithm

An algorithm to be used for bulk encryption. This specification includes the key size of this algorithm, whether it is a block, stream, or AEAD cipher, the block size of the cipher (if appropriate), and the lengths of explicit and implicit initialization vectors (or nonces).

MAC algorithm

An algorithm to be used for message authentication. This specification includes the size of the value returned by the MAC algorithm.

compression algorithm

An algorithm to be used for data compression. This specification must include all information the algorithm requires to do compression.

master secret

A 48-byte secret shared between the two peers in the connection.

client random

A 32-byte value provided by the client.

server random

A 32-byte value provided by the server.

These parameters are defined in the presentation language as:

```
enum { server, client } ConnectionEnd;
```

```
enum { tls_prf_sha256 } PRFAlgorithm;
```

```
enum { null, rc4, 3des, aes }  
      BulkCipherAlgorithm;
```

```
enum { stream, block, aead } CipherType;
```

```
enum { null, hmac_md5, hmac_sha1, hmac_sha256,  
      hmac_sha384, hmac_sha512 } MACAlgorithm;
```

```
enum { null(0), (255) } CompressionMethod;
```

```
/* The algorithms specified in CompressionMethod, PRFAlgorithm,  
   BulkCipherAlgorithm, and MACAlgorithm may be added to. */
```

```
struct {  
    ConnectionEnd          entity;  
    PRFAlgorithm            prf_algorithm;  
    BulkCipherAlgorithm    bulk_cipher_algorithm;  
    CipherType              cipher_type;  
    uint8                   enc_key_length;  
    uint8                   block_length;  
    uint8                   fixed_iv_length;  
    uint8                   record_iv_length;  
    MACAlgorithm            mac_algorithm;  
    uint8                   mac_length;  
    uint8                   mac_key_length;  
    CompressionMethod       compression_algorithm;  
    opaque                  master_secret[48];  
    opaque                  client_random[32];  
    opaque                  server_random[32];  
} SecurityParameters;
```

The record layer will use the security parameters to generate the following six items (some of which are not required by all ciphers, and are thus empty):

```
client write MAC key  
server write MAC key  
client write encryption key  
server write encryption key  
client write IV  
server write IV
```

The client write parameters are used by the server when receiving and processing records and vice versa. The algorithm used for generating these items from the security parameters is described in Section 6.3.

Once the security parameters have been set and the keys have been generated, the connection states can be instantiated by making them the current states. These current states **MUST** be updated for each record processed. Each connection state includes the following elements:

compression state

The current state of the compression algorithm.

cipher state

The current state of the encryption algorithm. This will consist of the scheduled key for that connection. For stream ciphers, this will also contain whatever state information is necessary to allow the stream to continue to encrypt or decrypt data.

MAC key

The MAC key for this connection, as generated above.

sequence number

Each connection state contains a sequence number, which is maintained separately for read and write states. The sequence number MUST be set to zero whenever a connection state is made the active state. Sequence numbers are of type uint64 and may not exceed $2^{64}-1$. Sequence numbers do not wrap. If a TLS implementation would need to wrap a sequence number, it must renegotiate instead. A sequence number is incremented after each record: specifically, the first record transmitted under a particular connection state MUST use sequence number 0.

6.2. Record Layer

The TLS record layer receives uninterpreted data from higher layers in non-empty blocks of arbitrary size.

6.2.1. Fragmentation

The record layer fragments information blocks into TLSPlaintext records carrying data in chunks of 2^{14} bytes or less. Client message boundaries are not preserved in the record layer (i.e., multiple client messages of the same ContentType MAY be coalesced into a single TLSPlaintext record, or a single message MAY be fragmented across several records).

```

struct {
    uint8 major;
    uint8 minor;
} ProtocolVersion;

enum {
    change_cipher_spec(20), alert(21), handshake(22),
    application_data(23), (255)
} ContentType;

struct {
    ContentType type;
    ProtocolVersion version;
    uint16 length;
    opaque fragment[TLSPlaintext.length];
} TLSPlaintext;

```

type

The higher-level protocol used to process the enclosed fragment.

version

The version of the protocol being employed. This document describes TLS Version 1.2, which uses the version { 3, 3 }. The version value 3.3 is historical, deriving from the use of {3, 1} for TLS 1.0. (See Appendix A.1.) Note that a client that supports multiple versions of TLS may not know what version will be employed before it receives the ServerHello. See Appendix E for discussion about what record layer version number should be employed for ClientHello.

length

The length (in bytes) of the following TLSPlaintext.fragment. The length MUST NOT exceed 2^{14} .

fragment

The application data. This data is transparent and treated as an independent block to be dealt with by the higher-level protocol specified by the type field.

Implementations MUST NOT send zero-length fragments of Handshake, Alert, or ChangeCipherSpec content types. Zero-length fragments of Application data MAY be sent as they are potentially useful as a traffic analysis countermeasure.

Note: Data of different TLS record layer content types MAY be interleaved. Application data is generally of lower precedence for transmission than other content types. However, records MUST be delivered to the network in the same order as they are protected by the record layer. Recipients MUST receive and process interleaved application layer traffic during handshakes subsequent to the first one on a connection.

6.2.2. Record Compression and Decompression

All records are compressed using the compression algorithm defined in the current session state. There is always an active compression algorithm; however, initially it is defined as CompressionMethod.null. The compression algorithm translates a TLSPlaintext structure into a TLSCompressed structure. Compression functions are initialized with default state information whenever a connection state is made active. [RFC3749] describes compression algorithms for TLS.

Compression must be lossless and may not increase the content length by more than 1024 bytes. If the decompression function encounters a TLSCompressed.fragment that would decompress to a length in excess of 2^{14} bytes, it MUST report a fatal decompression failure error.

```
struct {
    ContentType type;          /* same as TLSPlaintext.type */
    ProtocolVersion version; /* same as TLSPlaintext.version */
    uint16 length;
    opaque fragment[TLSCompressed.length];
} TLSCompressed;
```

length

The length (in bytes) of the following TLSCompressed.fragment.
The length MUST NOT exceed $2^{14} + 1024$.

fragment

The compressed form of TLSPlaintext.fragment.

Note: A CompressionMethod.null operation is an identity operation;
no fields are altered.

Implementation note: Decompression functions are responsible for
ensuring that messages cannot cause internal buffer overflows.

6.2.3. Record Payload Protection

The encryption and MAC functions translate a TLSCompressed
structure into a TLSCiphertext. The decryption functions reverse
the process. The MAC of the record also includes a sequence
number so that missing, extra, or repeated messages are
detectable.

```
struct {
    ContentType type;
    ProtocolVersion version;
    uint16 length;
    select (SecurityParameters.cipher_type) {
        case stream: GenericStreamCipher;
        case block:  GenericBlockCipher;
        case aead:   GenericAEADCipher;
    } fragment;
} TLSCiphertext;
```

type

The type field is identical to TLSCompressed.type.

version

The version field is identical to TLSCompressed.version.

length

The length (in bytes) of the following TLSCiphertext.fragment.
The length MUST NOT exceed $2^{14} + 2048$.

fragment

The encrypted form of `TLSCompressed.fragment`, with the MAC.

6.2.3.1. Null or Standard Stream Cipher

Stream ciphers (including `BulkCipherAlgorithm.null`; see Appendix A.6) convert `TLSCompressed.fragment` structures to and from stream `TLSCiphertext.fragment` structures.

```
stream-ciphered struct {
    opaque content[TLSCompressed.length];
    opaque MAC[SecurityParameters.mac_length];
} GenericStreamCipher;
```

The MAC is generated as:

```
MAC(MAC_write_key, seq_num +
    TLSCompressed.type +
    TLSCompressed.version +
    TLSCompressed.length +
    TLSCompressed.fragment);
```

where "+" denotes concatenation.

`seq_num`

The sequence number for this record.

`MAC`

The MAC algorithm specified by `SecurityParameters.mac_algorithm`.

Note that the MAC is computed before encryption. The stream cipher encrypts the entire block, including the MAC. For stream ciphers that do not use a synchronization vector (such as RC4), the stream cipher state from the end of one record is simply used on the subsequent packet. If the cipher suite is `TLS_NULL_WITH_NULL_NULL`, encryption consists of the identity operation (i.e., the data is not encrypted, and the MAC size is zero, implying that no MAC is used). For both null and stream ciphers, `TLSCiphertext.length` is `TLSCompressed.length` plus `SecurityParameters.mac_length`.

6.2.3.2. CBC Block Cipher

For block ciphers (such as 3DES or AES), the encryption and MAC functions convert `TLSCompressed.fragment` structures to and from block `TLSCiphertext.fragment` structures.

```
struct {
    opaque IV[SecurityParameters.record_iv_length];
    block-ciphered struct {
        opaque content[TLSCompressed.length];
        opaque MAC[SecurityParameters.mac_length];
        uint8 padding[GenericBlockCipher.padding_length];
        uint8 padding_length;
    };
} GenericBlockCipher;
```

The MAC is generated as described in Section 6.2.3.1.

IV

The Initialization Vector (IV) SHOULD be chosen at random, and MUST be unpredictable. Note that in versions of TLS prior to 1.1, there was no IV field, and the last ciphertext block of the previous record (the "CBC residue") was used as the IV. This was changed to prevent the attacks described in [CBCATT]. For block ciphers, the IV length is of length `SecurityParameters.record_iv_length`, which is equal to the `SecurityParameters.block_size`.

padding

Padding that is added to force the length of the plaintext to be an integral multiple of the block cipher's block length. The padding MAY be any length up to 255 bytes, as long as it results in the `TLSCiphertext.length` being an integral multiple of the block length. Lengths longer than necessary might be desirable to frustrate attacks on a protocol that are based on analysis of the lengths of exchanged messages. Each `uint8` in the padding data vector MUST be filled with the padding length value. The receiver MUST check this padding and MUST use the `bad_record_mac` alert to indicate padding errors.

padding_length

The padding length MUST be such that the total size of the `GenericBlockCipher` structure is a multiple of the cipher's block length. Legal values range from zero to 255, inclusive. This length specifies the length of the padding field exclusive of the `padding_length` field itself.

The encrypted data length (`TLSCiphertext.length`) is one more than the sum of `SecurityParameters.block_length`, `TLSCompressed.length`, `SecurityParameters.mac_length`, and `padding_length`.

Example: If the block length is 8 bytes, the content length (`TLSCompressed.length`) is 61 bytes, and the MAC length is 20 bytes, then the length before padding is 82 bytes (this does not include the

IV. Thus, the padding length modulo 8 must be equal to 6 in order to make the total length an even multiple of 8 bytes (the block length). The padding length can be 6, 14, 22, and so on, through 254. If the padding length were the minimum necessary, 6, the padding would be 6 bytes, each containing the value 6. Thus, the last 8 octets of the GenericBlockCipher before block encryption would be xx 06 06 06 06 06 06 06, where xx is the last octet of the MAC.

Note: With block ciphers in CBC mode (Cipher Block Chaining), it is critical that the entire plaintext of the record be known before any ciphertext is transmitted. Otherwise, it is possible for the attacker to mount the attack described in [CBCATT].

Implementation note: Canvel et al. [CBCTIME] have demonstrated a timing attack on CBC padding based on the time required to compute the MAC. In order to defend against this attack, implementations MUST ensure that record processing time is essentially the same whether or not the padding is correct. In general, the best way to do this is to compute the MAC even if the padding is incorrect, and only then reject the packet. For instance, if the pad appears to be incorrect, the implementation might assume a zero-length pad and then compute the MAC. This leaves a small timing channel, since MAC performance depends to some extent on the size of the data fragment, but it is not believed to be large enough to be exploitable, due to the large block size of existing MACs and the small size of the timing signal.

6.2.3.3. AEAD Ciphers

For AEAD [AEAD] ciphers (such as [CCM] or [GCM]), the AEAD function converts TLSCompressed.fragment structures to and from AEAD TLSCiphertext.fragment structures.

```
struct {
    opaque nonce_explicit[SecurityParameters.record_iv_length];
    aead-ciphered struct {
        opaque content[TLSCompressed.length];
    };
} GenericAEADCipher;
```

AEAD ciphers take as input a single key, a nonce, a plaintext, and "additional data" to be included in the authentication check, as described in Section 2.1 of [AEAD]. The key is either the client_write_key or the server_write_key. No MAC key is used.

Each AEAD cipher suite MUST specify how the nonce supplied to the AEAD operation is constructed, and what is the length of the GenericAEADCipher.nonce_explicit part. In many cases, it is

appropriate to use the partially implicit nonce technique described in Section 3.2.1 of [AEAD]; with `record_iv_length` being the length of the explicit part. In this case, the implicit part SHOULD be derived from `key_block` as `client_write_iv` and `server_write_iv` (as described in Section 6.3), and the explicit part is included in `GenericAEADCipher.nonce_explicit`.

The plaintext is the `TLSCompressed.fragment`.

The additional authenticated data, which we denote as `additional_data`, is defined as follows:

```
additional_data = seq_num + TLSCompressed.type +
                  TLSCompressed.version + TLSCompressed.length;
```

where "+" denotes concatenation.

The `aead_output` consists of the ciphertext output by the AEAD encryption operation. The length will generally be larger than `TLSCompressed.length`, but by an amount that varies with the AEAD cipher. Since the ciphers might incorporate padding, the amount of overhead could vary with different `TLSCompressed.length` values. Each AEAD cipher MUST NOT produce an expansion of greater than 1024 bytes. Symbolically,

```
AEADEncrypted = AEAD-Encrypt(write_key, nonce, plaintext,
                             additional_data)
```

In order to decrypt and verify, the cipher takes as input the key, nonce, the "additional_data", and the `AEADEncrypted` value. The output is either the plaintext or an error indicating that the decryption failed. There is no separate integrity check. That is:

```
TLSCompressed.fragment = AEAD-Decrypt(write_key, nonce,
                                       AEADEncrypted,
                                       additional_data)
```

If the decryption fails, a fatal `bad_record_mac` alert MUST be generated.

6.3. Key Calculation

The Record Protocol requires an algorithm to generate keys required by the current connection state (see Appendix A.6) from the security parameters provided by the handshake protocol.

The master secret is expanded into a sequence of secure bytes, which is then split to a client write MAC key, a server write MAC key, a client write encryption key, and a server write encryption key. Each of these is generated from the byte sequence in that order. Unused values are empty. Some AEAD ciphers may additionally require a client write IV and a server write IV (see Section 6.2.3.3).

When keys and MAC keys are generated, the master secret is used as an entropy source.

To generate the key material, compute

```
key_block = PRF(SecurityParameters.master_secret,
                "key expansion",
                SecurityParameters.server_random +
                SecurityParameters.client_random);
```

until enough output has been generated. Then, the key_block is partitioned as follows:

```
client_write_MAC_key[SecurityParameters.mac_key_length]
server_write_MAC_key[SecurityParameters.mac_key_length]
client_write_key[SecurityParameters.enc_key_length]
server_write_key[SecurityParameters.enc_key_length]
client_write_IV[SecurityParameters.fixed_iv_length]
server_write_IV[SecurityParameters.fixed_iv_length]
```

Currently, the client_write_IV and server_write_IV are only generated for implicit nonce techniques as described in Section 3.2.1 of [AEAD].

Implementation note: The currently defined cipher suite which requires the most material is AES_256_CBC_SHA256. It requires 2 x 32 byte keys and 2 x 32 byte MAC keys, for a total 128 bytes of key material.

7. The TLS Handshaking Protocols

TLS has three subprotocols that are used to allow peers to agree upon security parameters for the record layer, to authenticate themselves, to instantiate negotiated security parameters, and to report error conditions to each other.

The Handshake Protocol is responsible for negotiating a session, which consists of the following items:

session identifier

An arbitrary byte sequence chosen by the server to identify an active or resumable session state.

peer certificate

X509v3 [PKIX] certificate of the peer. This element of the state may be null.

compression method

The algorithm used to compress data prior to encryption.

cipher spec

Specifies the pseudorandom function (PRF) used to generate keying material, the bulk data encryption algorithm (such as null, AES, etc.) and the MAC algorithm (such as HMAC-SHA1). It also defines cryptographic attributes such as the `mac_length`. (See Appendix A.6 for formal definition.)

master secret

48-byte secret shared between the client and server.

is resumable

A flag indicating whether the session can be used to initiate new connections.

These items are then used to create security parameters for use by the record layer when protecting application data. Many connections can be instantiated using the same session through the resumption feature of the TLS Handshake Protocol.

7.1. Change Cipher Spec Protocol

The change cipher spec protocol exists to signal transitions in ciphering strategies. The protocol consists of a single message, which is encrypted and compressed under the current (not the pending) connection state. The message consists of a single byte of value 1.

```
struct {  
    enum { change_cipher_spec(1), (255) } type;  
} ChangeCipherSpec;
```

The ChangeCipherSpec message is sent by both the client and the server to notify the receiving party that subsequent records will be protected under the newly negotiated CipherSpec and keys. Reception of this message causes the receiver to instruct the record layer to immediately copy the read pending state into the read current state. Immediately after sending this message, the sender MUST instruct the record layer to make the write pending state the write active state.

(See Section 6.1.) The ChangeCipherSpec message is sent during the handshake after the security parameters have been agreed upon, but before the verifying Finished message is sent.

Note: If a rehandshake occurs while data is flowing on a connection, the communicating parties may continue to send data using the old CipherSpec. However, once the ChangeCipherSpec has been sent, the new CipherSpec MUST be used. The first side to send the ChangeCipherSpec does not know that the other side has finished computing the new keying material (e.g., if it has to perform a time-consuming public key operation). Thus, a small window of time, during which the recipient must buffer the data, MAY exist. In practice, with modern machines this interval is likely to be fairly short.

7.2. Alert Protocol

One of the content types supported by the TLS record layer is the alert type. Alert messages convey the severity of the message (warning or fatal) and a description of the alert. Alert messages with a level of fatal result in the immediate termination of the connection. In this case, other connections corresponding to the session may continue, but the session identifier MUST be invalidated, preventing the failed session from being used to establish new connections. Like other messages, alert messages are encrypted and compressed, as specified by the current connection state.

```
enum { warning(1), fatal(2), (255) } AlertLevel;
```

```
enum {  
    close_notify(0),  
    unexpected_message(10),  
    bad_record_mac(20),  
    decryption_failed_RESERVED(21),  
    record_overflow(22),  
    decompression_failure(30),  
    handshake_failure(40),  
    no_certificate_RESERVED(41),  
    bad_certificate(42),  
    unsupported_certificate(43),  
    certificate_revoked(44),  
    certificate_expired(45),  
    certificate_unknown(46),  
    illegal_parameter(47),  
    unknown_ca(48),  
    access_denied(49),  
    decode_error(50),  
    decrypt_error(51),  
}
```

```
    export_restriction_RESERVED(60),
    protocol_version(70),
    insufficient_security(71),
    internal_error(80),
    user_canceled(90),
    no_renegotiation(100),
    unsupported_extension(110),
    (255)
} AlertDescription;

struct {
    AlertLevel level;
    AlertDescription description;
} Alert;
```

7.2.1. Closure Alerts

The client and the server must share knowledge that the connection is ending in order to avoid a truncation attack. Either party may initiate the exchange of closing messages.

`close_notify`

This message notifies the recipient that the sender will not send any more messages on this connection. Note that as of TLS 1.1, failure to properly close a connection no longer requires that a session not be resumed. This is a change from TLS 1.0 to conform with widespread implementation practice.

Either party may initiate a close by sending a `close_notify` alert. Any data received after a closure alert is ignored.

Unless some other fatal alert has been transmitted, each party is required to send a `close_notify` alert before closing the write side of the connection. The other party **MUST** respond with a `close_notify` alert of its own and close down the connection immediately, discarding any pending writes. It is not required for the initiator of the close to wait for the responding `close_notify` alert before closing the read side of the connection.

If the application protocol using TLS provides that any data may be carried over the underlying transport after the TLS connection is closed, the TLS implementation must receive the responding `close_notify` alert before indicating to the application layer that the TLS connection has ended. If the application protocol will not transfer any additional data, but will only close the underlying transport connection, then the implementation **MAY** choose to close the transport without waiting for the responding `close_notify`. No part

of this standard should be taken to dictate the manner in which a usage profile for TLS manages its data transport, including when connections are opened or closed.

Note: It is assumed that closing a connection reliably delivers pending data before destroying the transport.

7.2.2. Error Alerts

Error handling in the TLS Handshake protocol is very simple. When an error is detected, the detecting party sends a message to the other party. Upon transmission or receipt of a fatal alert message, both parties immediately close the connection. Servers and clients **MUST** forget any session-identifiers, keys, and secrets associated with a failed connection. Thus, any connection terminated with a fatal alert **MUST NOT** be resumed.

Whenever an implementation encounters a condition which is defined as a fatal alert, it **MUST** send the appropriate alert prior to closing the connection. For all errors where an alert level is not explicitly specified, the sending party **MAY** determine at its discretion whether to treat this as a fatal error or not. If the implementation chooses to send an alert but intends to close the connection immediately afterwards, it **MUST** send that alert at the fatal alert level.

If an alert with a level of warning is sent and received, generally the connection can continue normally. If the receiving party decides not to proceed with the connection (e.g., after having received a `no_renegotiation` alert that it is not willing to accept), it **SHOULD** send a fatal alert to terminate the connection. Given this, the sending party cannot, in general, know how the receiving party will behave. Therefore, warning alerts are not very useful when the sending party wants to continue the connection, and thus are sometimes omitted. For example, if a peer decides to accept an expired certificate (perhaps after confirming this with the user) and wants to continue the connection, it would not generally send a `certificate_expired` alert.

The following error alerts are defined:

`unexpected_message`

An inappropriate message was received. This alert is always fatal and should never be observed in communication between proper implementations.

bad_record_mac

This alert is returned if a record is received with an incorrect MAC. This alert also MUST be returned if an alert is sent because a TLSCiphertext decrypted in an invalid way: either it wasn't an even multiple of the block length, or its padding values, when checked, weren't correct. This message is always fatal and should never be observed in communication between proper implementations (except when messages were corrupted in the network).

decryption_failed_RESERVED

This alert was used in some earlier versions of TLS, and may have permitted certain attacks against the CBC mode [CBCATT]. It MUST NOT be sent by compliant implementations.

record_overflow

A TLSCiphertext record was received that had a length more than $2^{14}+2048$ bytes, or a record decrypted to a TLSCompressed record with more than $2^{14}+1024$ bytes. This message is always fatal and should never be observed in communication between proper implementations (except when messages were corrupted in the network).

decompression_failure

The decompression function received improper input (e.g., data that would expand to excessive length). This message is always fatal and should never be observed in communication between proper implementations.

handshake_failure

Reception of a handshake_failure alert message indicates that the sender was unable to negotiate an acceptable set of security parameters given the options available. This is a fatal error.

no_certificate_RESERVED

This alert was used in SSLv3 but not any version of TLS. It MUST NOT be sent by compliant implementations.

bad_certificate

A certificate was corrupt, contained signatures that did not verify correctly, etc.

unsupported_certificate

A certificate was of an unsupported type.

certificate_revoked

A certificate was revoked by its signer.

certificate_expired

A certificate has expired or is not currently valid.

certificate_unknown

Some other (unspecified) issue arose in processing the certificate, rendering it unacceptable.

illegal_parameter

A field in the handshake was out of range or inconsistent with other fields. This message is always fatal.

unknown_ca

A valid certificate chain or partial chain was received, but the certificate was not accepted because the CA certificate could not be located or couldn't be matched with a known, trusted CA. This message is always fatal.

access_denied

A valid certificate was received, but when access control was applied, the sender decided not to proceed with negotiation. This message is always fatal.

decode_error

A message could not be decoded because some field was out of the specified range or the length of the message was incorrect. This message is always fatal and should never be observed in communication between proper implementations (except when messages were corrupted in the network).

decrypt_error

A handshake cryptographic operation failed, including being unable to correctly verify a signature or validate a Finished message. This message is always fatal.

export_restriction_RESERVED

This alert was used in some earlier versions of TLS. It MUST NOT be sent by compliant implementations.

protocol_version

The protocol version the client has attempted to negotiate is recognized but not supported. (For example, old protocol versions might be avoided for security reasons.) This message is always fatal.

insufficient_security

Returned instead of `handshake_failure` when a negotiation has failed specifically because the server requires ciphers more secure than those supported by the client. This message is always fatal.

internal_error

An internal error unrelated to the peer or the correctness of the protocol (such as a memory allocation failure) makes it impossible to continue. This message is always fatal.

user_canceled

This handshake is being canceled for some reason unrelated to a protocol failure. If the user cancels an operation after the handshake is complete, just closing the connection by sending a `close_notify` is more appropriate. This alert should be followed by a `close_notify`. This message is generally a warning.

no_renegotiation

Sent by the client in response to a hello request or by the server in response to a client hello after initial handshaking. Either of these would normally lead to renegotiation; when that is not appropriate, the recipient should respond with this alert. At that point, the original requester can decide whether to proceed with the connection. One case where this would be appropriate is where a server has spawned a process to satisfy a request; the process might receive security parameters (key length, authentication, etc.) at startup, and it might be difficult to communicate changes to these parameters after that point. This message is always a warning.

unsupported_extension

sent by clients that receive an extended server hello containing an extension that they did not put in the corresponding client hello. This message is always fatal.

New Alert values are assigned by IANA as described in Section 12.

7.3. Handshake Protocol Overview

The cryptographic parameters of the session state are produced by the TLS Handshake Protocol, which operates on top of the TLS record layer. When a TLS client and server first start communicating, they agree on a protocol version, select cryptographic algorithms, optionally authenticate each other, and use public-key encryption techniques to generate shared secrets.

The TLS Handshake Protocol involves the following steps:

- Exchange hello messages to agree on algorithms, exchange random values, and check for session resumption.
- Exchange the necessary cryptographic parameters to allow the client and server to agree on a premaster secret.
- Exchange certificates and cryptographic information to allow the client and server to authenticate themselves.
- Generate a master secret from the premaster secret and exchanged random values.
- Provide security parameters to the record layer.
- Allow the client and server to verify that their peer has calculated the same security parameters and that the handshake occurred without tampering by an attacker.

Note that higher layers should not be overly reliant on whether TLS always negotiates the strongest possible connection between two peers. There are a number of ways in which a man-in-the-middle attacker can attempt to make two entities drop down to the least secure method they support. The protocol has been designed to minimize this risk, but there are still attacks available: for example, an attacker could block access to the port a secure service runs on, or attempt to get the peers to negotiate an unauthenticated connection. The fundamental rule is that higher levels must be cognizant of what their security requirements are and never transmit information over a channel less secure than what they require. The TLS protocol is secure in that any cipher suite offers its promised level of security: if you negotiate 3DES with a 1024-bit RSA key exchange with a host whose certificate you have verified, you can expect to be that secure.

These goals are achieved by the handshake protocol, which can be summarized as follows: The client sends a ClientHello message to which the server must respond with a ServerHello message, or else a fatal error will occur and the connection will fail. The ClientHello and ServerHello are used to establish security enhancement capabilities between client and server. The ClientHello and ServerHello establish the following attributes: Protocol Version, Session ID, Cipher Suite, and Compression Method. Additionally, two random values are generated and exchanged: ClientHello.random and ServerHello.random.

The actual key exchange uses up to four messages: the server Certificate, the ServerKeyExchange, the client Certificate, and the ClientKeyExchange. New key exchange methods can be created by specifying a format for these messages and by defining the use of the messages to allow the client and server to agree upon a shared secret. This secret **MUST** be quite long; currently defined key exchange methods exchange secrets that range from 46 bytes upwards.

Following the hello messages, the server will send its certificate in a Certificate message if it is to be authenticated. Additionally, a ServerKeyExchange message may be sent, if it is required (e.g., if the server has no certificate, or if its certificate is for signing only). If the server is authenticated, it may request a certificate from the client, if that is appropriate to the cipher suite selected. Next, the server will send the ServerHelloDone message, indicating that the hello-message phase of the handshake is complete. The server will then wait for a client response. If the server has sent a CertificateRequest message, the client **MUST** send the Certificate message. The ClientKeyExchange message is now sent, and the content of that message will depend on the public key algorithm selected between the ClientHello and the ServerHello. If the client has sent a certificate with signing ability, a digitally-signed CertificateVerify message is sent to explicitly verify possession of the private key in the certificate.

At this point, a ChangeCipherSpec message is sent by the client, and the client copies the pending Cipher Spec into the current Cipher Spec. The client then immediately sends the Finished message under the new algorithms, keys, and secrets. In response, the server will send its own ChangeCipherSpec message, transfer the pending to the current Cipher Spec, and send its Finished message under the new Cipher Spec. At this point, the handshake is complete, and the client and server may begin to exchange application layer data. (See flow chart below.) Application data **MUST NOT** be sent prior to the completion of the first handshake (before a cipher suite other than TLS_NULL_WITH_NULL_NULL is established).

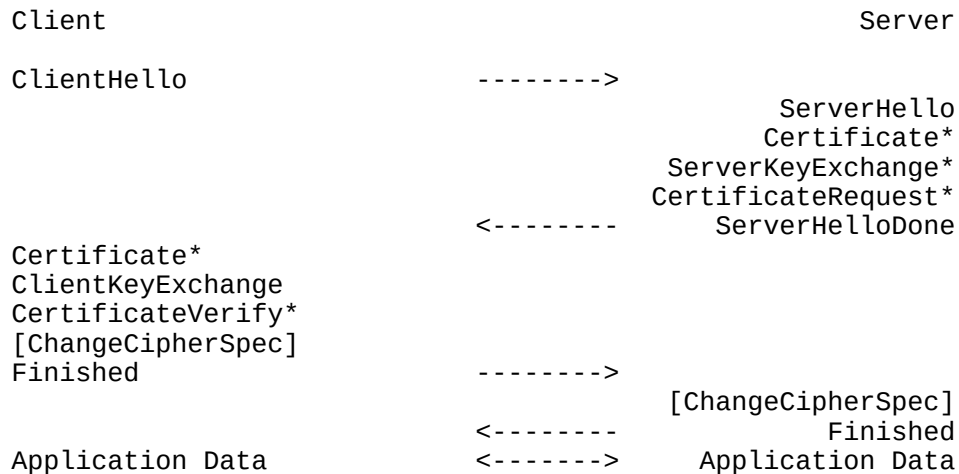


Figure 1. Message flow for a full handshake

* Indicates optional or situation-dependent messages that are not always sent.

Note: To help avoid pipeline stalls, ChangeCipherSpec is an independent TLS protocol content type, and is not actually a TLS handshake message.

When the client and server decide to resume a previous session or duplicate an existing session (instead of negotiating new security parameters), the message flow is as follows:

The client sends a ClientHello using the Session ID of the session to be resumed. The server then checks its session cache for a match. If a match is found, and the server is willing to re-establish the connection under the specified session state, it will send a ServerHello with the same Session ID value. At this point, both client and server MUST send ChangeCipherSpec messages and proceed directly to Finished messages. Once the re-establishment is complete, the client and server MAY begin to exchange application layer data. (See flow chart below.) If a Session ID match is not found, the server generates a new session ID, and the TLS client and server perform a full handshake.

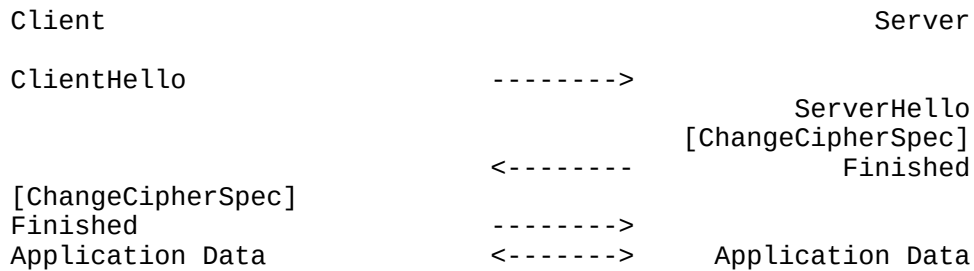


Figure 2. Message flow for an abbreviated handshake

The contents and significance of each message will be presented in detail in the following sections.

7.4. Handshake Protocol

The TLS Handshake Protocol is one of the defined higher-level clients of the TLS Record Protocol. This protocol is used to negotiate the secure attributes of a session. Handshake messages are supplied to the TLS record layer, where they are encapsulated within one or more TLSPlaintext structures, which are processed and transmitted as specified by the current active session state.

```

enum {
    hello_request(0), client_hello(1), server_hello(2),
    certificate(11), server_key_exchange (12),
    certificate_request(13), server_hello_done(14),
    certificate_verify(15), client_key_exchange(16),
    finished(20), (255)
} HandshakeType;

struct {
    HandshakeType msg_type; /* handshake type */
    uint24 length; /* bytes in message */
    select (HandshakeType) {
        case hello_request: HelloRequest;
        case client_hello: ClientHello;
        case server_hello: ServerHello;
        case certificate: Certificate;
        case server_key_exchange: ServerKeyExchange;
        case certificate_request: CertificateRequest;
        case server_hello_done: ServerHelloDone;
        case certificate_verify: CertificateVerify;
        case client_key_exchange: ClientKeyExchange;
        case finished: Finished;
    } body;
} Handshake;
  
```

The handshake protocol messages are presented below in the order they MUST be sent; sending handshake messages in an unexpected order results in a fatal error. Unneeded handshake messages can be omitted, however. Note one exception to the ordering: the Certificate message is used twice in the handshake (from server to client, then from client to server), but described only in its first position. The one message that is not bound by these ordering rules is the HelloRequest message, which can be sent at any time, but which SHOULD be ignored by the client if it arrives in the middle of a handshake.

New handshake message types are assigned by IANA as described in Section 12.

7.4.1. Hello Messages

The hello phase messages are used to exchange security enhancement capabilities between the client and server. When a new session begins, the record layer's connection state encryption, hash, and compression algorithms are initialized to null. The current connection state is used for renegotiation messages.

7.4.1.1. Hello Request

When this message will be sent:

The HelloRequest message MAY be sent by the server at any time.

Meaning of this message:

HelloRequest is a simple notification that the client should begin the negotiation process anew. In response, the client should send a ClientHello message when convenient. This message is not intended to establish which side is the client or server but merely to initiate a new negotiation. Servers SHOULD NOT send a HelloRequest immediately upon the client's initial connection. It is the client's job to send a ClientHello at that time.

This message will be ignored by the client if the client is currently negotiating a session. This message MAY be ignored by the client if it does not wish to renegotiate a session, or the client may, if it wishes, respond with a no_renegotiation alert. Since handshake messages are intended to have transmission precedence over application data, it is expected that the negotiation will begin before no more than a few records are received from the client. If the server sends a HelloRequest but does not receive a ClientHello in response, it may close the connection with a fatal alert.

After sending a HelloRequest, servers SHOULD NOT repeat the request until the subsequent handshake negotiation is complete.

Structure of this message:

```
struct { } HelloRequest;
```

This message MUST NOT be included in the message hashes that are maintained throughout the handshake and used in the Finished messages and the certificate verify message.

7.4.1.2. Client Hello

When this message will be sent:

When a client first connects to a server, it is required to send the ClientHello as its first message. The client can also send a ClientHello in response to a HelloRequest or on its own initiative in order to renegotiate the security parameters in an existing connection.

Structure of this message:

The ClientHello message includes a random structure, which is used later in the protocol.

```
struct {  
    uint32  gmt_unix_time;  
    opaque  random_bytes[28];  
} Random;
```

gmt_unix_time

The current time and date in standard UNIX 32-bit format (seconds since the midnight starting Jan 1, 1970, UTC, ignoring leap seconds) according to the sender's internal clock. Clocks are not required to be set correctly by the basic TLS protocol; higher-level or application protocols may define additional requirements. Note that, for historical reasons, the data element is named using GMT, the predecessor of the current worldwide time base, UTC.

random_bytes

28 bytes generated by a secure random number generator.

The ClientHello message includes a variable-length session identifier. If not empty, the value identifies a session between the same client and server whose security parameters the client wishes to reuse. The session identifier MAY be from an earlier connection,

this connection, or from another currently active connection. The second option is useful if the client only wishes to update the random structures and derived values of a connection, and the third option makes it possible to establish several independent secure connections without repeating the full handshake protocol. These independent connections may occur sequentially or simultaneously; a SessionID becomes valid when the handshake negotiating it completes with the exchange of Finished messages and persists until it is removed due to aging or because a fatal error was encountered on a connection associated with the session. The actual contents of the SessionID are defined by the server.

```
opaque SessionID<0..32>;
```

Warning: Because the SessionID is transmitted without encryption or immediate MAC protection, servers MUST NOT place confidential information in session identifiers or let the contents of fake session identifiers cause any breach of security. (Note that the content of the handshake as a whole, including the SessionID, is protected by the Finished messages exchanged at the end of the handshake.)

The cipher suite list, passed from the client to the server in the ClientHello message, contains the combinations of cryptographic algorithms supported by the client in order of the client's preference (favorite choice first). Each cipher suite defines a key exchange algorithm, a bulk encryption algorithm (including secret key length), a MAC algorithm, and a PRF. The server will select a cipher suite or, if no acceptable choices are presented, return a handshake failure alert and close the connection. If the list contains cipher suites the server does not recognize, support, or wish to use, the server MUST ignore those cipher suites, and process the remaining ones as usual.

```
uint8 CipherSuite[2]; /* Cryptographic suite selector */
```

The ClientHello includes a list of compression algorithms supported by the client, ordered according to the client's preference.

```
enum { null(0), (255) } CompressionMethod;
```



```
struct {  
    ProtocolVersion client_version;  
    Random random;  
    SessionID session_id;  
    CipherSuite cipher_suites<2..2^16-2>;  
    CompressionMethod compression_methods<1..2^8-1>;  
    select (extensions_present) {  
        case false:  
            struct {};  
        case true:  
            Extension extensions<0..2^16-1>;  
    };  
} ClientHello;
```

TLS allows extensions to follow the `compression_methods` field in an extensions block. The presence of extensions can be detected by determining whether there are bytes following the `compression_methods` at the end of the `ClientHello`. Note that this method of detecting optional data differs from the normal TLS method of having a variable-length field, but it is used for compatibility with TLS before extensions were defined.

`client_version`

The version of the TLS protocol by which the client wishes to communicate during this session. This SHOULD be the latest (highest valued) version supported by the client. For this version of the specification, the version will be 3.3 (see Appendix E for details about backward compatibility).

`random`

A client-generated random structure.

`session_id`

The ID of a session the client wishes to use for this connection. This field is empty if no `session_id` is available, or if the client wishes to generate new security parameters.

`cipher_suites`

This is a list of the cryptographic options supported by the client, with the client's first preference first. If the `session_id` field is not empty (implying a session resumption request), this vector MUST include at least the `cipher_suite` from that session. Values are defined in Appendix A.5.

`compression_methods`

This is a list of the compression methods supported by the client, sorted by client preference. If the `session_id` field is not empty (implying a session resumption request), it MUST include the

compression_method from that session. This vector MUST contain, and all implementations MUST support, CompressionMethod.null. Thus, a client and server will always be able to agree on a compression method.

extensions

Clients MAY request extended functionality from servers by sending data in the extensions field. The actual "Extension" format is defined in Section 7.4.1.4.

In the event that a client requests additional functionality using extensions, and this functionality is not supplied by the server, the client MAY abort the handshake. A server MUST accept ClientHello messages both with and without the extensions field, and (as for all other messages) it MUST check that the amount of data in the message precisely matches one of these formats; if not, then it MUST send a fatal "decode_error" alert.

After sending the ClientHello message, the client waits for a ServerHello message. Any handshake message returned by the server, except for a HelloRequest, is treated as a fatal error.

7.4.1.3. Server Hello

When this message will be sent:

The server will send this message in response to a ClientHello message when it was able to find an acceptable set of algorithms. If it cannot find such a match, it will respond with a handshake failure alert.

Structure of this message:

```
struct {  
    ProtocolVersion server_version;  
    Random random;  
    SessionID session_id;  
    CipherSuite cipher_suite;  
    CompressionMethod compression_method;  
    select (extensions_present) {  
        case false:  
            struct {};  
        case true:  
            Extension extensions<0..2^16-1>;  
    };  
} ServerHello;
```

The presence of extensions can be detected by determining whether there are bytes following the `compression_method` field at the end of the `ServerHello`.

`server_version`

This field will contain the lower of that suggested by the client in the client hello and the highest supported by the server. For this version of the specification, the version is 3.3. (See Appendix E for details about backward compatibility.)

`random`

This structure is generated by the server and MUST be independently generated from the `ClientHello.random`.

`session_id`

This is the identity of the session corresponding to this connection. If the `ClientHello.session_id` was non-empty, the server will look in its session cache for a match. If a match is found and the server is willing to establish the new connection using the specified session state, the server will respond with the same value as was supplied by the client. This indicates a resumed session and dictates that the parties must proceed directly to the Finished messages. Otherwise, this field will contain a different value identifying the new session. The server may return an empty `session_id` to indicate that the session will not be cached and therefore cannot be resumed. If a session is resumed, it must be resumed using the same cipher suite it was originally negotiated with. Note that there is no requirement that the server resume any session even if it had formerly provided a `session_id`. Clients MUST be prepared to do a full negotiation -- including negotiating new cipher suites -- during any handshake.

`cipher_suite`

The single cipher suite selected by the server from the list in `ClientHello.cipher_suites`. For resumed sessions, this field is the value from the state of the session being resumed.

`compression_method`

The single compression algorithm selected by the server from the list in `ClientHello.compression_methods`. For resumed sessions, this field is the value from the resumed session state.

`extensions`

A list of extensions. Note that only extensions offered by the client can appear in the server's list.

7.4.1.4. Hello Extensions

The extension format is:

```
struct {  
    ExtensionType extension_type;  
    opaque extension_data<0..2^16-1>;  
} Extension;  
  
enum {  
    signature_algorithms(13), (65535)  
} ExtensionType;
```

Here:

- "extension_type" identifies the particular extension type.
- "extension_data" contains information specific to the particular extension type.

The initial set of extensions is defined in a companion document [TLSEXT]. The list of extension types is maintained by IANA as described in Section 12.

An extension type MUST NOT appear in the ServerHello unless the same extension type appeared in the corresponding ClientHello. If a client receives an extension type in ServerHello that it did not request in the associated ClientHello, it MUST abort the handshake with an unsupported_extension fatal alert.

Nonetheless, "server-oriented" extensions may be provided in the future within this framework. Such an extension (say, of type x) would require the client to first send an extension of type x in a ClientHello with empty extension_data to indicate that it supports the extension type. In this case, the client is offering the capability to understand the extension type, and the server is taking the client up on its offer.

When multiple extensions of different types are present in the ClientHello or ServerHello messages, the extensions MAY appear in any order. There MUST NOT be more than one extension of the same type.

Finally, note that extensions can be sent both when starting a new session and when requesting session resumption. Indeed, a client that requests session resumption does not in general know whether the server will accept this request, and therefore it SHOULD send the same extensions as it would send if it were not attempting resumption.

In general, the specification of each extension type needs to describe the effect of the extension both during full handshake and session resumption. Most current TLS extensions are relevant only when a session is initiated: when an older session is resumed, the server does not process these extensions in Client Hello, and does not include them in Server Hello. However, some extensions may specify different behavior during session resumption.

There are subtle (and not so subtle) interactions that may occur in this protocol between new features and existing features which may result in a significant reduction in overall security. The following considerations should be taken into account when designing new extensions:

- Some cases where a server does not agree to an extension are error conditions, and some are simply refusals to support particular features. In general, error alerts should be used for the former, and a field in the server extension response for the latter.
- Extensions should, as far as possible, be designed to prevent any attack that forces use (or non-use) of a particular feature by manipulation of handshake messages. This principle should be followed regardless of whether the feature is believed to cause a security problem.

Often the fact that the extension fields are included in the inputs to the Finished message hashes will be sufficient, but extreme care is needed when the extension changes the meaning of messages sent in the handshake phase. Designers and implementors should be aware of the fact that until the handshake has been authenticated, active attackers can modify messages and insert, remove, or replace extensions.

- It would be technically possible to use extensions to change major aspects of the design of TLS; for example the design of cipher suite negotiation. This is not recommended; it would be more appropriate to define a new version of TLS -- particularly since the TLS handshake algorithms have specific protection against version rollback attacks based on the version number, and the possibility of version rollback should be a significant consideration in any major design change.

7.4.1.4.1. Signature Algorithms

The client uses the "signature_algorithms" extension to indicate to the server which signature/hash algorithm pairs may be used in digital signatures. The "extension_data" field of this extension contains a "supported_signature_algorithms" value.

```

enum {
    none(0), md5(1), sha1(2), sha224(3), sha256(4), sha384(5),
    sha512(6), (255)
} HashAlgorithm;

enum { anonymous(0), rsa(1), dsa(2), ecdsa(3), (255) }
SignatureAlgorithm;

struct {
    HashAlgorithm hash;
    SignatureAlgorithm signature;
} SignatureAndHashAlgorithm;

SignatureAndHashAlgorithm
supported_signature_algorithms<2..2^16-2>;

```

Each SignatureAndHashAlgorithm value lists a single hash/signature pair that the client is willing to verify. The values are indicated in descending order of preference.

Note: Because not all signature algorithms and hash algorithms may be accepted by an implementation (e.g., DSA with SHA-1, but not SHA-256), algorithms here are listed in pairs.

hash

This field indicates the hash algorithm which may be used. The values indicate support for unhashed data, MD5 [MD5], SHA-1, SHA-224, SHA-256, SHA-384, and SHA-512 [SHS], respectively. The "none" value is provided for future extensibility, in case of a signature algorithm which does not require hashing before signing.

signature

This field indicates the signature algorithm that may be used. The values indicate anonymous signatures, RSASSA-PKCS1-v1_5 [PKCS1] and DSA [DSS], and ECDSA [ECDSA], respectively. The "anonymous" value is meaningless in this context but used in Section 7.4.3. It MUST NOT appear in this extension.

The semantics of this extension are somewhat complicated because the cipher suite indicates permissible signature algorithms but not hash algorithms. Sections 7.4.2 and 7.4.3 describe the appropriate rules.

If the client supports only the default hash and signature algorithms (listed in this section), it MAY omit the signature_algorithms extension. If the client does not support the default algorithms, or supports other hash and signature algorithms (and it is willing to use them for verifying messages sent by the server, i.e., server certificates and server key exchange), it MUST send the

signature_algorithms extension, listing the algorithms it is willing to accept.

If the client does not send the signature_algorithms extension, the server MUST do the following:

- If the negotiated key exchange algorithm is one of (RSA, DHE_RSA, DH_RSA, RSA_PSK, ECDH_RSA, ECDHE_RSA), behave as if client had sent the value {sha1,rsa}.
- If the negotiated key exchange algorithm is one of (DHE_DSS, DH_DSS), behave as if the client had sent the value {sha1,dsa}.
- If the negotiated key exchange algorithm is one of (ECDH_ECDSA, ECDHE_ECDSA), behave as if the client had sent value {sha1,ecdsa}.

Note: this is a change from TLS 1.1 where there are no explicit rules, but as a practical matter one can assume that the peer supports MD5 and SHA-1.

Note: this extension is not meaningful for TLS versions prior to 1.2. Clients MUST NOT offer it if they are offering prior versions. However, even if clients do offer it, the rules specified in [TLSEXT] require servers to ignore extensions they do not understand.

Servers MUST NOT send this extension. TLS servers MUST support receiving this extension.

When performing session resumption, this extension is not included in Server Hello, and the server ignores the extension in Client Hello (if present).

7.4.2. Server Certificate

When this message will be sent:

The server MUST send a Certificate message whenever the agreed-upon key exchange method uses certificates for authentication (this includes all key exchange methods defined in this document except DH_anon). This message will always immediately follow the ServerHello message.

Meaning of this message:

This message conveys the server's certificate chain to the client.

The certificate MUST be appropriate for the negotiated cipher suite's key exchange algorithm and any negotiated extensions.

Structure of this message:

```
opaque ASN.1Cert<1..2^24-1>;

struct {
    ASN.1Cert certificate_list<0..2^24-1>;
} Certificate;
```

certificate_list

This is a sequence (chain) of certificates. The sender's certificate MUST come first in the list. Each following certificate MUST directly certify the one preceding it. Because certificate validation requires that root keys be distributed independently, the self-signed certificate that specifies the root certificate authority MAY be omitted from the chain, under the assumption that the remote end must already possess it in order to validate it in any case.

The same message type and structure will be used for the client's response to a certificate request message. Note that a client MAY send no certificates if it does not have an appropriate certificate to send in response to the server's authentication request.

Note: PKCS #7 [PKCS7] is not used as the format for the certificate vector because PKCS #6 [PKCS6] extended certificates are not used. Also, PKCS #7 defines a SET rather than a SEQUENCE, making the task of parsing the list more difficult.

The following rules apply to the certificates sent by the server:

- The certificate type MUST be X.509v3, unless explicitly negotiated otherwise (e.g., [TLSPGP]).
- The end entity certificate's public key (and associated restrictions) MUST be compatible with the selected key exchange algorithm.

Key Exchange Alg. Certificate Key Type

RSA	RSA public key; the certificate MUST allow the
RSA_PSK	key to be used for encryption (the
	keyEncipherment bit MUST be set if the key
	usage extension is present).
	Note: RSA_PSK is defined in [TLSPSK].

DHE_RSA
ECDHE_RSA

RSA public key; the certificate MUST allow the key to be used for signing (the `digitalSignature` bit MUST be set if the key usage extension is present) with the signature scheme and hash algorithm that will be employed in the server key exchange message. Note: ECDHE_RSA is defined in [TLSECC].

DHE_DSS

DSA public key; the certificate MUST allow the key to be used for signing with the hash algorithm that will be employed in the server key exchange message.

DH_DSS
DH_RSA

Diffie-Hellman public key; the `keyAgreement` bit MUST be set if the key usage extension is present.

ECDH_ECDSA
ECDH_RSA

ECDH-capable public key; the public key MUST use a curve and point format supported by the client, as described in [TLSECC].

ECDHE_ECDSA

ECDSA-capable public key; the certificate MUST allow the key to be used for signing with the hash algorithm that will be employed in the server key exchange message. The public key MUST use a curve and point format supported by the client, as described in [TLSECC].

- The "server_name" and "trusted_ca_keys" extensions [TLSEXT] are used to guide certificate selection.

If the client provided a "signature_algorithms" extension, then all certificates provided by the server MUST be signed by a hash/signature algorithm pair that appears in that extension. Note that this implies that a certificate containing a key for one signature algorithm MAY be signed using a different signature algorithm (for instance, an RSA key signed with a DSA key). This is a departure from TLS 1.1, which required that the algorithms be the same. Note that this also implies that the DH_DSS, DH_RSA, ECDH_ECDSA, and ECDH_RSA key exchange algorithms do not restrict the algorithm used to sign the certificate. Fixed DH certificates MAY be signed with any hash/signature algorithm pair appearing in the extension. The names DH_DSS, DH_RSA, ECDH_ECDSA, and ECDH_RSA are historical.

If the server has multiple certificates, it chooses one of them based on the above-mentioned criteria (in addition to other criteria, such as transport layer endpoint, local configuration and preferences, etc.). If the server has a single certificate, it SHOULD attempt to validate that it meets these criteria.

Note that there are certificates that use algorithms and/or algorithm combinations that cannot be currently used with TLS. For example, a certificate with RSASSA-PSS signature key (id-RSASSA-PSS OID in SubjectPublicKeyInfo) cannot be used because TLS defines no corresponding signature algorithm.

As cipher suites that specify new key exchange methods are specified for the TLS protocol, they will imply the certificate format and the required encoded keying information.

7.4.3. Server Key Exchange Message

When this message will be sent:

This message will be sent immediately after the server Certificate message (or the ServerHello message, if this is an anonymous negotiation).

The ServerKeyExchange message is sent by the server only when the server Certificate message (if sent) does not contain enough data to allow the client to exchange a premaster secret. This is true for the following key exchange methods:

- DHE_DSS
- DHE_RSA
- DH_anon

It is not legal to send the ServerKeyExchange message for the following key exchange methods:

- RSA
- DH_DSS
- DH_RSA

Other key exchange algorithms, such as those defined in [TLSECC], MUST specify whether the ServerKeyExchange message is sent or not; and if the message is sent, its contents.

Meaning of this message:

This message conveys cryptographic information to allow the client to communicate the premaster secret: a Diffie-Hellman public key with which the client can complete a key exchange (with the result being the premaster secret) or a public key for some other algorithm.

Structure of this message:

```
enum { dhe_dss, dhe_rsa, dh_anon, rsa, dh_dss, dh_rsa
      /* may be extended, e.g., for ECDH -- see [TLSECC] */
      } KeyExchangeAlgorithm;
```

```
struct {
    opaque dh_p<1..216-1>;
    opaque dh_g<1..216-1>;
    opaque dh_Ys<1..216-1>;
} ServerDHPParams;      /* Ephemeral DH parameters */
```

dh_p
The prime modulus used for the Diffie-Hellman operation.

dh_g
The generator used for the Diffie-Hellman operation.

dh_Ys
The server's Diffie-Hellman public value ($g^X \bmod p$).

```

struct {
    select (KeyExchangeAlgorithm) {
        case dh_anon:
            ServerDHParams params;
        case dhe_dss:
        case dhe_rsa:
            ServerDHParams params;
            digitally-signed struct {
                opaque client_random[32];
                opaque server_random[32];
                ServerDHParams params;
            } signed_params;
        case rsa:
        case dh_dss:
        case dh_rsa:
            struct {} ;
        /* message is omitted for rsa, dh_dss, and dh_rsa */
        /* may be extended, e.g., for ECDH -- see [TLSECC] */
    };
} ServerKeyExchange;

```

params

The server's key exchange parameters.

signed_params

For non-anonymous key exchanges, a signature over the server's key exchange parameters.

If the client has offered the "signature_algorithms" extension, the signature algorithm and hash algorithm MUST be a pair listed in that extension. Note that there is a possibility for inconsistencies here. For instance, the client might offer DHE_DSS key exchange but omit any DSA pairs from its "signature_algorithms" extension. In order to negotiate correctly, the server MUST check any candidate cipher suites against the "signature_algorithms" extension before selecting them. This is somewhat inelegant but is a compromise designed to minimize changes to the original cipher suite design.

In addition, the hash and signature algorithms MUST be compatible with the key in the server's end-entity certificate. RSA keys MAY be used with any permitted hash algorithm, subject to restrictions in the certificate, if any.

Because DSA signatures do not contain any secure indication of hash algorithm, there is a risk of hash substitution if multiple hashes may be used with any key. Currently, DSA [DSS] may only be used with SHA-1. Future revisions of DSS [DSS-3] are expected to allow the use of other digest algorithms with DSA, as well as guidance as to which

digest algorithms should be used with each key size. In addition, future revisions of [PKIX] may specify mechanisms for certificates to indicate which digest algorithms are to be used with DSA.

As additional cipher suites are defined for TLS that include new key exchange algorithms, the server key exchange message will be sent if and only if the certificate type associated with the key exchange algorithm does not provide enough information for the client to exchange a premaster secret.

7.4.4. Certificate Request

When this message will be sent:

A non-anonymous server can optionally request a certificate from the client, if appropriate for the selected cipher suite. This message, if sent, will immediately follow the ServerKeyExchange message (if it is sent; otherwise, this message follows the server's Certificate message).

Structure of this message:

```
enum {
    rsa_sign(1), dss_sign(2), rsa_fixed_dh(3), dss_fixed_dh(4),
    rsa_ephemeral_dh_RESERVED(5), dss_ephemeral_dh_RESERVED(6),
    fortezza_dms_RESERVED(20), (255)
} ClientCertificateType;

opaque DistinguishedName<1..2^16-1>;

struct {
    ClientCertificateType certificate_types<1..2^8-1>;
    SignatureAndHashAlgorithm
        supported_signature_algorithms<2^16-1>;
    DistinguishedName certificate_authorities<0..2^16-1>;
} CertificateRequest;
```

certificate_types

A list of the types of certificate types that the client may offer.

rsa_sign	a certificate containing an RSA key
dss_sign	a certificate containing a DSA key
rsa_fixed_dh	a certificate containing a static DH key.
dss_fixed_dh	a certificate containing a static DH key

supported_signature_algorithms

A list of the hash/signature algorithm pairs that the server is able to verify, listed in descending order of preference.

certificate_authorities

A list of the distinguished names [X501] of acceptable certificate_authorities, represented in DER-encoded format. These distinguished names may specify a desired distinguished name for a root CA or for a subordinate CA; thus, this message can be used to describe known roots as well as a desired authorization space. If the certificate_authorities list is empty, then the client MAY send any certificate of the appropriate ClientCertificateType, unless there is some external arrangement to the contrary.

The interaction of the certificate_types and supported_signature_algorithms fields is somewhat complicated. certificate_types has been present in TLS since SSLv3, but was somewhat underspecified. Much of its functionality is superseded by supported_signature_algorithms. The following rules apply:

- Any certificates provided by the client MUST be signed using a hash/signature algorithm pair found in supported_signature_algorithms.
- The end-entity certificate provided by the client MUST contain a key that is compatible with certificate_types. If the key is a signature key, it MUST be usable with some hash/signature algorithm pair in supported_signature_algorithms.
- For historical reasons, the names of some client certificate types include the algorithm used to sign the certificate. For example, in earlier versions of TLS, rsa_fixed_dh meant a certificate signed with RSA and containing a static DH key. In TLS 1.2, this functionality has been obsoleted by the supported_signature_algorithms, and the certificate type no longer restricts the algorithm used to sign the certificate. For example, if the server sends dss_fixed_dh certificate type and {{sha1, dsa}, {sha1, rsa}} signature types, the client MAY reply with a certificate containing a static DH key, signed with RSA-SHA1.

New ClientCertificateType values are assigned by IANA as described in Section 12.

Note: Values listed as RESERVED may not be used. They were used in SSLv3.

Note: It is a fatal handshake_failure alert for an anonymous server to request client authentication.

7.4.5. Server Hello Done

When this message will be sent:

The ServerHelloDone message is sent by the server to indicate the end of the ServerHello and associated messages. After sending this message, the server will wait for a client response.

Meaning of this message:

This message means that the server is done sending messages to support the key exchange, and the client can proceed with its phase of the key exchange.

Upon receipt of the ServerHelloDone message, the client SHOULD verify that the server provided a valid certificate, if required, and check that the server hello parameters are acceptable.

Structure of this message:

```
struct { } ServerHelloDone;
```

7.4.6. Client Certificate

When this message will be sent:

This is the first message the client can send after receiving a ServerHelloDone message. This message is only sent if the server requests a certificate. If no suitable certificate is available, the client MUST send a certificate message containing no certificates. That is, the certificate_list structure has a length of zero. If the client does not send any certificates, the server MAY at its discretion either continue the handshake without client authentication, or respond with a fatal handshake_failure alert. Also, if some aspect of the certificate chain was unacceptable (e.g., it was not signed by a known, trusted CA), the server MAY at its discretion either continue the handshake (considering the client unauthenticated) or send a fatal alert.

Client certificates are sent using the Certificate structure defined in Section 7.4.2.

Meaning of this message:

This message conveys the client's certificate chain to the server; the server will use it when verifying the CertificateVerify message (when the client authentication is based on signing) or calculating the premaster secret (for non-ephemeral Diffie-Hellman). The certificate MUST be appropriate for the negotiated cipher suite's key exchange algorithm, and any negotiated extensions.

In particular:

- The certificate type MUST be X.509v3, unless explicitly negotiated otherwise (e.g., [TLSPGP]).
- The end-entity certificate's public key (and associated restrictions) has to be compatible with the certificate types listed in CertificateRequest:

Client Cert. Type	Certificate Key Type
rsa_sign	RSA public key; the certificate MUST allow the key to be used for signing with the signature scheme and hash algorithm that will be employed in the certificate verify message.
dss_sign	DSA public key; the certificate MUST allow the key to be used for signing with the hash algorithm that will be employed in the certificate verify message.
ecdsa_sign	ECDSA-capable public key; the certificate MUST allow the key to be used for signing with the hash algorithm that will be employed in the certificate verify message; the public key MUST use a curve and point format supported by the server.
rsa_fixed_dh dss_fixed_dh	Diffie-Hellman public key; MUST use the same parameters as server's key.
rsa_fixed_ecdh ecdsa_fixed_ecdh	ECDH-capable public key; MUST use the same curve as the server's key, and MUST use a point format supported by the server.
- If the certificate_authorities list in the certificate request message was non-empty, one of the certificates in the certificate chain SHOULD be issued by one of the listed CAs.	

- The certificates MUST be signed using an acceptable hash/signature algorithm pair, as described in Section 7.4.4. Note that this relaxes the constraints on certificate-signing algorithms found in prior versions of TLS.

Note that, as with the server certificate, there are certificates that use algorithms/algorithm combinations that cannot be currently used with TLS.

7.4.7. Client Key Exchange Message

When this message will be sent:

This message is always sent by the client. It MUST immediately follow the client certificate message, if it is sent. Otherwise, it MUST be the first message sent by the client after it receives the ServerHelloDone message.

Meaning of this message:

With this message, the premaster secret is set, either by direct transmission of the RSA-encrypted secret or by the transmission of Diffie-Hellman parameters that will allow each side to agree upon the same premaster secret.

When the client is using an ephemeral Diffie-Hellman exponent, then this message contains the client's Diffie-Hellman public value. If the client is sending a certificate containing a static DH exponent (i.e., it is doing `fixed_dh` client authentication), then this message MUST be sent but MUST be empty.

Structure of this message:

The choice of messages depends on which key exchange method has been selected. See Section 7.4.3 for the `KeyExchangeAlgorithm` definition.

```

struct {
    select (KeyExchangeAlgorithm) {
        case rsa:
            EncryptedPreMasterSecret;
        case dhe_dss:
        case dhe_rsa:
        case dh_dss:
        case dh_rsa:
        case dh_anon:
            ClientDiffieHellmanPublic;
    } exchange_keys;
} ClientKeyExchange;

```

7.4.7.1. RSA-Encrypted Premaster Secret Message

Meaning of this message:

If RSA is being used for key agreement and authentication, the client generates a 48-byte premaster secret, encrypts it using the public key from the server's certificate, and sends the result in an encrypted premaster secret message. This structure is a variant of the ClientKeyExchange message and is not a message in itself.

Structure of this message:

```

struct {
    ProtocolVersion client_version;
    opaque random[46];
} PreMasterSecret;

```

client_version

The latest (newest) version supported by the client. This is used to detect version rollback attacks.

random

46 securely-generated random bytes.

```

struct {
    public-key-encrypted PreMasterSecret pre_master_secret;
} EncryptedPreMasterSecret;

```

pre_master_secret

This random value is generated by the client and is used to generate the master secret, as specified in Section 8.1.

Note: The version number in the PreMasterSecret is the version offered by the client in the ClientHello.client_version, not the version negotiated for the connection. This feature is designed to prevent rollback attacks. Unfortunately, some old implementations use the negotiated version instead, and therefore checking the version number may lead to failure to interoperate with such incorrect client implementations.

Client implementations MUST always send the correct version number in PreMasterSecret. If ClientHello.client_version is TLS 1.1 or higher, server implementations MUST check the version number as described in the note below. If the version number is TLS 1.0 or earlier, server implementations SHOULD check the version number, but MAY have a configuration option to disable the check. Note that if the check fails, the PreMasterSecret SHOULD be randomized as described below.

Note: Attacks discovered by Bleichenbacher [BLEI] and Klima et al. [KPR03] can be used to attack a TLS server that reveals whether a particular message, when decrypted, is properly PKCS#1 formatted, contains a valid PreMasterSecret structure, or has the correct version number.

As described by Klima [KPR03], these vulnerabilities can be avoided by treating incorrectly formatted message blocks and/or mismatched version numbers in a manner indistinguishable from correctly formatted RSA blocks. In other words:

1. Generate a string R of 46 random bytes
2. Decrypt the message to recover the plaintext M
3. If the PKCS#1 padding is not correct, or the length of message M is not exactly 48 bytes:
 pre_master_secret = ClientHello.client_version || R
 else If ClientHello.client_version <= TLS 1.0, and version number check is explicitly disabled:
 pre_master_secret = M
 else:
 pre_master_secret = ClientHello.client_version || M[2..47]

Note that explicitly constructing the pre_master_secret with the ClientHello.client_version produces an invalid master_secret if the client has sent the wrong version in the original pre_master_secret.

An alternative approach is to treat a version number mismatch as a PKCS-1 formatting error and randomize the premaster secret completely:

1. Generate a string R of 48 random bytes
2. Decrypt the message to recover the plaintext M
3. If the PKCS#1 padding is not correct, or the length of message M is not exactly 48 bytes:
 - pre_master_secret = R
 - else If ClientHello.client_version <= TLS 1.0, and version number check is explicitly disabled:
 - premaster secret = M
 - else If M[0..1] != ClientHello.client_version:
 - premaster secret = R
 - else:
 - premaster secret = M

Although no practical attacks against this construction are known, Klima et al. [KPR03] describe some theoretical attacks, and therefore the first construction described is RECOMMENDED.

In any case, a TLS server MUST NOT generate an alert if processing an RSA-encrypted premaster secret message fails, or the version number is not as expected. Instead, it MUST continue the handshake with a randomly generated premaster secret. It may be useful to log the real cause of failure for troubleshooting purposes; however, care must be taken to avoid leaking the information to an attacker (through, e.g., timing, log files, or other channels.)

The RSAES-OAEP encryption scheme defined in [PKCS1] is more secure against the Bleichenbacher attack. However, for maximal compatibility with earlier versions of TLS, this specification uses the RSAES-PKCS1-v1_5 scheme. No variants of the Bleichenbacher attack are known to exist provided that the above recommendations are followed.

Implementation note: Public-key-encrypted data is represented as an opaque vector $\langle 0..2^{16}-1 \rangle$ (see Section 4.7). Thus, the RSA-encrypted PreMasterSecret in a ClientKeyExchange is preceded by two length bytes. These bytes are redundant in the case of RSA because the EncryptedPreMasterSecret is the only data in the ClientKeyExchange and its length can therefore be unambiguously determined. The SSLv3 specification was not clear about the encoding of public-key-encrypted data, and therefore many SSLv3 implementations do not include the length bytes -- they encode the RSA-encrypted data directly in the ClientKeyExchange message.

This specification requires correct encoding of the EncryptedPreMasterSecret complete with length bytes. The resulting PDU is incompatible with many SSLv3 implementations. Implementors

upgrading from SSLv3 MUST modify their implementations to generate and accept the correct encoding. Implementors who wish to be compatible with both SSLv3 and TLS should make their implementation's behavior dependent on the protocol version.

Implementation note: It is now known that remote timing-based attacks on TLS are possible, at least when the client and server are on the same LAN. Accordingly, implementations that use static RSA keys MUST use RSA blinding or some other anti-timing technique, as described in [TIMING].

7.4.7.2. Client Diffie-Hellman Public Value

Meaning of this message:

This structure conveys the client's Diffie-Hellman public value (Yc) if it was not already included in the client's certificate. The encoding used for Yc is determined by the enumerated PublicValueEncoding. This structure is a variant of the client key exchange message, and not a message in itself.

Structure of this message:

```
enum { implicit, explicit } PublicValueEncoding;
```

```
implicit
```

If the client has sent a certificate which contains a suitable Diffie-Hellman key (for fixed_dh client authentication), then Yc is implicit and does not need to be sent again. In this case, the client key exchange message will be sent, but it MUST be empty.

```
explicit
```

Yc needs to be sent.

```
struct {  
    select (PublicValueEncoding) {  
        case implicit: struct { };  
        case explicit: opaque dh_Yc<1..2^16-1>;  
    } dh_public;  
} ClientDiffieHellmanPublic;
```

```
dh_Yc
```

The client's Diffie-Hellman public value (Yc).

7.4.8. Certificate Verify

When this message will be sent:

This message is used to provide explicit verification of a client certificate. This message is only sent following a client certificate that has signing capability (i.e., all certificates except those containing fixed Diffie-Hellman parameters). When sent, it MUST immediately follow the client key exchange message.

Structure of this message:

```
struct {  
    digitally-signed struct {  
        opaque handshake_messages[handshake_messages_length];  
    }  
} CertificateVerify;
```

Here handshake_messages refers to all handshake messages sent or received, starting at client hello and up to, but not including, this message, including the type and length fields of the handshake messages. This is the concatenation of all the Handshake structures (as defined in Section 7.4) exchanged thus far. Note that this requires both sides to either buffer the messages or compute running hashes for all potential hash algorithms up to the time of the CertificateVerify computation. Servers can minimize this computation cost by offering a restricted set of digest algorithms in the CertificateRequest message.

The hash and signature algorithms used in the signature MUST be one of those present in the supported_signature_algorithms field of the CertificateRequest message. In addition, the hash and signature algorithms MUST be compatible with the key in the client's end-entity certificate. RSA keys MAY be used with any permitted hash algorithm, subject to restrictions in the certificate, if any.

Because DSA signatures do not contain any secure indication of hash algorithm, there is a risk of hash substitution if multiple hashes may be used with any key. Currently, DSA [DSS] may only be used with SHA-1. Future revisions of DSS [DSS-3] are expected to allow the use of other digest algorithms with DSA, as well as guidance as to which digest algorithms should be used with each key size. In addition, future revisions of [PKIX] may specify mechanisms for certificates to indicate which digest algorithms are to be used with DSA.

7.4.9. Finished

When this message will be sent:

A Finished message is always sent immediately after a change cipher spec message to verify that the key exchange and authentication processes were successful. It is essential that a change cipher spec message be received between the other handshake messages and the Finished message.

Meaning of this message:

The Finished message is the first one protected with the just negotiated algorithms, keys, and secrets. Recipients of Finished messages MUST verify that the contents are correct. Once a side has sent its Finished message and received and validated the Finished message from its peer, it may begin to send and receive application data over the connection.

Structure of this message:

```
struct {  
    opaque verify_data[verify_data_length];  
} Finished;  
  
verify_data  
    PRF(master_secret, finished_label, Hash(handshake_messages))  
    [0..verify_data_length-1];  
  
finished_label  
    For Finished messages sent by the client, the string  
    "client finished". For Finished messages sent by the server,  
    the string "server finished".
```

Hash denotes a Hash of the handshake messages. For the PRF defined in Section 5, the Hash MUST be the Hash used as the basis for the PRF. Any cipher suite which defines a different PRF MUST also define the Hash to use in the Finished computation.

In previous versions of TLS, the verify_data was always 12 octets long. In the current version of TLS, it depends on the cipher suite. Any cipher suite which does not explicitly specify verify_data_length has a verify_data_length equal to 12. This includes all existing cipher suites. Note that this representation has the same encoding as with previous versions. Future cipher suites MAY specify other lengths but such length MUST be at least 12 bytes.

handshake_messages

All of the data from all messages in this handshake (not including any HelloRequest messages) up to, but not including, this message. This is only data visible at the handshake layer and does not include record layer headers. This is the concatenation of all the Handshake structures as defined in Section 7.4, exchanged thus far.

It is a fatal error if a Finished message is not preceded by a ChangeCipherSpec message at the appropriate point in the handshake.

The value handshake_messages includes all handshake messages starting at ClientHello up to, but not including, this Finished message. This may be different from handshake_messages in Section 7.4.8 because it would include the CertificateVerify message (if sent). Also, the handshake_messages for the Finished message sent by the client will be different from that for the Finished message sent by the server, because the one that is sent second will include the prior one.

Note: ChangeCipherSpec messages, alerts, and any other record types are not handshake messages and are not included in the hash computations. Also, HelloRequest messages are omitted from handshake hashes.

8. Cryptographic Computations

In order to begin connection protection, the TLS Record Protocol requires specification of a suite of algorithms, a master secret, and the client and server random values. The authentication, encryption, and MAC algorithms are determined by the cipher_suite selected by the server and revealed in the ServerHello message. The compression algorithm is negotiated in the hello messages, and the random values are exchanged in the hello messages. All that remains is to calculate the master secret.

8.1. Computing the Master Secret

For all key exchange methods, the same algorithm is used to convert the pre_master_secret into the master_secret. The pre_master_secret should be deleted from memory once the master_secret has been computed.

```
master_secret = PRF(pre_master_secret, "master secret",
                    ClientHello.random + ServerHello.random)
                    [0..47];
```

The master secret is always exactly 48 bytes in length. The length of the premaster secret will vary depending on key exchange method.

8.1.1. RSA

When RSA is used for server authentication and key exchange, a 48-byte `pre_master_secret` is generated by the client, encrypted under the server's public key, and sent to the server. The server uses its private key to decrypt the `pre_master_secret`. Both parties then convert the `pre_master_secret` into the `master_secret`, as specified above.

8.1.2. Diffie-Hellman

A conventional Diffie-Hellman computation is performed. The negotiated key (Z) is used as the `pre_master_secret`, and is converted into the `master_secret`, as specified above. Leading bytes of Z that contain all zero bits are stripped before it is used as the `pre_master_secret`.

Note: Diffie-Hellman parameters are specified by the server and may be either ephemeral or contained within the server's certificate.

9. Mandatory Cipher Suites

In the absence of an application profile standard specifying otherwise, a TLS-compliant application MUST implement the cipher suite `TLS_RSA_WITH_AES_128_CBC_SHA` (see Appendix A.5 for the definition).

10. Application Data Protocol

Application data messages are carried by the record layer and are fragmented, compressed, and encrypted based on the current connection state. The messages are treated as transparent data to the record layer.

11. Security Considerations

Security issues are discussed throughout this memo, especially in Appendices D, E, and F.

12. IANA Considerations

This document uses several registries that were originally created in [TLS1.1]. IANA has updated these to reference this document. The registries and their allocation policies (unchanged from [TLS1.1]) are listed below.

- TLS ClientCertificateType Identifiers Registry: Future values in the range 0-63 (decimal) inclusive are assigned via Standards Action [RFC2434]. Values in the range 64-223 (decimal) inclusive are assigned via Specification Required [RFC2434]. Values from 224-255 (decimal) inclusive are reserved for Private Use [RFC2434].
- TLS Cipher Suite Registry: Future values with the first byte in the range 0-191 (decimal) inclusive are assigned via Standards Action [RFC2434]. Values with the first byte in the range 192-254 (decimal) are assigned via Specification Required [RFC2434]. Values with the first byte 255 (decimal) are reserved for Private Use [RFC2434].
- This document defines several new HMAC-SHA256-based cipher suites, whose values (in Appendix A.5) have been allocated from the TLS Cipher Suite registry.
- TLS ContentType Registry: Future values are allocated via Standards Action [RFC2434].
- TLS Alert Registry: Future values are allocated via Standards Action [RFC2434].
- TLS HandshakeType Registry: Future values are allocated via Standards Action [RFC2434].

This document also uses a registry originally created in [RFC4366]. IANA has updated it to reference this document. The registry and its allocation policy (unchanged from [RFC4366]) is listed below:

- TLS ExtensionType Registry: Future values are allocated via IETF Consensus [RFC2434]. IANA has updated this registry to include the signature_algorithms extension and its corresponding value (see Section 7.4.1.4).

In addition, this document defines two new registries to be maintained by IANA:

- TLS SignatureAlgorithm Registry: The registry has been initially populated with the values described in Section 7.4.1.4.1. Future values in the range 0-63 (decimal) inclusive are assigned via Standards Action [RFC2434]. Values in the range 64-223 (decimal) inclusive are assigned via Specification Required [RFC2434]. Values from 224-255 (decimal) inclusive are reserved for Private Use [RFC2434].

- TLS HashAlgorithm Registry: The registry has been initially populated with the values described in Section 7.4.1.4.1. Future values in the range 0-63 (decimal) inclusive are assigned via Standards Action [RFC2434]. Values in the range 64-223 (decimal) inclusive are assigned via Specification Required [RFC2434]. Values from 224-255 (decimal) inclusive are reserved for Private Use [RFC2434].

This document also uses the TLS Compression Method Identifiers Registry, defined in [RFC3749]. IANA has allocated value 0 for the "null" compression method.

Appendix A. Protocol Data Structures and Constant Values

This section describes protocol types and constants.

A.1. Record Layer

```
struct {
    uint8 major;
    uint8 minor;
} ProtocolVersion;

ProtocolVersion version = { 3, 3 };    /* TLS v1.2*/

enum {
    change_cipher_spec(20), alert(21), handshake(22),
    application_data(23), (255)
} ContentType;

struct {
    ContentType type;
    ProtocolVersion version;
    uint16 length;
    opaque fragment[TLSPplaintext.length];
} TLSPplaintext;

struct {
    ContentType type;
    ProtocolVersion version;
    uint16 length;
    opaque fragment[TLSCompressed.length];
} TLSCompressed;

struct {
    ContentType type;
    ProtocolVersion version;
    uint16 length;
    select (SecurityParameters.cipher_type) {
        case stream: GenericStreamCipher;
        case block:  GenericBlockCipher;
        case aead:   GenericAEADCipher;
    } fragment;
} TLSCiphertext;

stream-ciphered struct {
    opaque content[TLSCompressed.length];
    opaque MAC[SecurityParameters.mac_length];
} GenericStreamCipher;
```

```

struct {
    opaque IV[SecurityParameters.record_iv_length];
    block-ciphered struct {
        opaque content[TLSCompressed.length];
        opaque MAC[SecurityParameters.mac_length];
        uint8 padding[GenericBlockCipher.padding_length];
        uint8 padding_length;
    };
} GenericBlockCipher;

struct {
    opaque nonce_explicit[SecurityParameters.record_iv_length];
    aead-ciphered struct {
        opaque content[TLSCompressed.length];
    };
} GenericAEADCipher;

```

A.2. Change Cipher Specs Message

```

struct {
    enum { change_cipher_spec(1), (255) } type;
} ChangeCipherSpec;

```

A.3. Alert Messages

```

enum { warning(1), fatal(2), (255) } AlertLevel;

enum {
    close_notify(0),
    unexpected_message(10),
    bad_record_mac(20),
    decryption_failed_RESERVED(21),
    record_overflow(22),
    decompression_failure(30),
    handshake_failure(40),
    no_certificate_RESERVED(41),
    bad_certificate(42),
    unsupported_certificate(43),
    certificate_revoked(44),
    certificate_expired(45),
    certificate_unknown(46),
    illegal_parameter(47),
    unknown_ca(48),
    access_denied(49),
    decode_error(50),
    decrypt_error(51),
    export_restriction_RESERVED(60),
    protocol_version(70),

```

```
        insufficient_security(71),
        internal_error(80),
        user_canceled(90),
        no_renegotiation(100),
        unsupported_extension(110),          /* new */
        (255)
    } AlertDescription;

    struct {
        AlertLevel level;
        AlertDescription description;
    } Alert;
}
```

A.4. Handshake Protocol

```
enum {
    hello_request(0), client_hello(1), server_hello(2),
    certificate(11), server_key_exchange (12),
    certificate_request(13), server_hello_done(14),
    certificate_verify(15), client_key_exchange(16),
    finished(20)
    (255)
} HandshakeType;

struct {
    HandshakeType msg_type;
    uint24 length;
    select (HandshakeType) {
        case hello_request:      HelloRequest;
        case client_hello:      ClientHello;
        case server_hello:      ServerHello;
        case certificate:        Certificate;
        case server_key_exchange: ServerKeyExchange;
        case certificate_request: CertificateRequest;
        case server_hello_done:  ServerHelloDone;
        case certificate_verify:  CertificateVerify;
        case client_key_exchange: ClientKeyExchange;
        case finished:           Finished;
    } body;
} Handshake;
```

A.4.1. Hello Messages

```
struct { } HelloRequest;

struct {
    uint32  gmt_unix_time;
    opaque random_bytes[28];
} Random;

opaque SessionID<0..32>;

uint8 CipherSuite[2];

enum { null(0), (255) } CompressionMethod;

struct {
    ProtocolVersion client_version;
    Random random;
    SessionID session_id;
    CipherSuite cipher_suites<2..2^16-2>;
    CompressionMethod compression_methods<1..2^8-1>;
    select (extensions_present) {
        case false:
            struct {};
        case true:
            Extension extensions<0..2^16-1>;
    };
} ClientHello;

struct {
    ProtocolVersion server_version;
    Random random;
    SessionID session_id;
    CipherSuite cipher_suite;
    CompressionMethod compression_method;
    select (extensions_present) {
        case false:
            struct {};
        case true:
            Extension extensions<0..2^16-1>;
    };
} ServerHello;

struct {
    ExtensionType extension_type;
    opaque extension_data<0..2^16-1>;
} Extension;
```

```

enum {
    signature_algorithms(13), (65535)
} ExtensionType;

enum{
    none(0), md5(1), sha1(2), sha224(3), sha256(4), sha384(5),
    sha512(6), (255)
} HashAlgorithm;
enum {
    anonymous(0), rsa(1), dsa(2), ecdsa(3), (255)
} SignatureAlgorithm;

struct {
    HashAlgorithm hash;
    SignatureAlgorithm signature;
} SignatureAndHashAlgorithm;

SignatureAndHashAlgorithm
supported_signature_algorithms<2..2^16-1>;

```

A.4.2. Server Authentication and Key Exchange Messages

```

opaque ASN.1Cert<2^24-1>;

struct {
    ASN.1Cert certificate_list<0..2^24-1>;
} Certificate;

enum { dhe_dss, dhe_rsa, dh_anon, rsa, dh_dss, dh_rsa
    /* may be extended, e.g., for ECDH -- see [TLSECC] */
} KeyExchangeAlgorithm;

struct {
    opaque dh_p<1..2^16-1>;
    opaque dh_g<1..2^16-1>;
    opaque dh_Ys<1..2^16-1>;
} ServerDHPParams; /* Ephemeral DH parameters */

```



```

struct {
    select (KeyExchangeAlgorithm) {
        case dh_anon:
            ServerDHParams params;
        case dhe_dss:
        case dhe_rsa:
            ServerDHParams params;
            digitally-signed struct {
                opaque client_random[32];
                opaque server_random[32];
                ServerDHParams params;
            } signed_params;
        case rsa:
        case dh_dss:
        case dh_rsa:
            struct {} ;
            /* message is omitted for rsa, dh_dss, and dh_rsa */
            /* may be extended, e.g., for ECDH -- see [TLSECC] */
    } ServerKeyExchange;

    enum {
        rsa_sign(1), dss_sign(2), rsa_fixed_dh(3), dss_fixed_dh(4),
        rsa_ephemeral_dh_RESERVED(5), dss_ephemeral_dh_RESERVED(6),
        fortaleza_dms_RESERVED(20),
        (255)
    } ClientCertificateType;

    opaque DistinguishedName<1..2^16-1>;

    struct {
        ClientCertificateType certificate_types<1..2^8-1>;
        DistinguishedName certificate_authorities<0..2^16-1>;
    } CertificateRequest;

    struct { } ServerHelloDone;

```

A.4.3. Client Authentication and Key Exchange Messages

```

struct {
    select (KeyExchangeAlgorithm) {
        case rsa:
            EncryptedPreMasterSecret;
        case dhe_dss:
        case dhe_rsa:
        case dh_dss:
        case dh_rsa:
        case dh_anon:
            ClientDiffieHellmanPublic;
    } exchange_keys;
} ClientKeyExchange;

struct {
    ProtocolVersion client_version;
    opaque random[46];
} PreMasterSecret;

struct {
    public-key-encrypted PreMasterSecret pre_master_secret;
} EncryptedPreMasterSecret;

enum { implicit, explicit } PublicValueEncoding;

struct {
    select (PublicValueEncoding) {
        case implicit: struct {};
        case explicit: opaque DH_Yc<1..2^16-1>;
    } dh_public;
} ClientDiffieHellmanPublic;

struct {
    digitally-signed struct {
        opaque handshake_messages[handshake_messages_length];
    }
} CertificateVerify;

```

A.4.4. Handshake Finalization Message

```

struct {
    opaque verify_data[verify_data_length];
} Finished;

```

A.5. The Cipher Suite

The following values define the cipher suite codes used in the ClientHello and ServerHello messages.

A cipher suite defines a cipher specification supported in TLS Version 1.2.

TLS_NULL_WITH_NULL_NULL is specified and is the initial state of a TLS connection during the first handshake on that channel, but MUST NOT be negotiated, as it provides no more protection than an unsecured connection.

```
CipherSuite TLS_NULL_WITH_NULL_NULL = { 0x00,0x00 };
```

The following CipherSuite definitions require that the server provide an RSA certificate that can be used for key exchange. The server may request any signature-capable certificate in the certificate request message.

```
CipherSuite TLS_RSA_WITH_NULL_MD5 = { 0x00,0x01 };
CipherSuite TLS_RSA_WITH_NULL_SHA = { 0x00,0x02 };
CipherSuite TLS_RSA_WITH_NULL_SHA256 = { 0x00,0x3B };
CipherSuite TLS_RSA_WITH_RC4_128_MD5 = { 0x00,0x04 };
CipherSuite TLS_RSA_WITH_RC4_128_SHA = { 0x00,0x05 };
CipherSuite TLS_RSA_WITH_3DES_EDE_CBC_SHA = { 0x00,0x0A };
CipherSuite TLS_RSA_WITH_AES_128_CBC_SHA = { 0x00,0x2F };
CipherSuite TLS_RSA_WITH_AES_256_CBC_SHA = { 0x00,0x35 };
CipherSuite TLS_RSA_WITH_AES_128_CBC_SHA256 = { 0x00,0x3C };
CipherSuite TLS_RSA_WITH_AES_256_CBC_SHA256 = { 0x00,0x3D };
```

The following cipher suite definitions are used for server-authenticated (and optionally client-authenticated) Diffie-Hellman. DH denotes cipher suites in which the server's certificate contains the Diffie-Hellman parameters signed by the certificate authority (CA). DHE denotes ephemeral Diffie-Hellman, where the Diffie-Hellman parameters are signed by a signature-capable certificate, which has been signed by the CA. The signing algorithm used by the server is specified after the DHE component of the CipherSuite name. The server can request any signature-capable certificate from the client for client authentication, or it may request a Diffie-Hellman certificate. Any Diffie-Hellman certificate provided by the client must use the parameters (group and generator) described by the server.

CipherSuite	TLS_DH_DSS_WITH_3DES_EDE_CBC_SHA	= { 0x00, 0x0D };
CipherSuite	TLS_DH_RSA_WITH_3DES_EDE_CBC_SHA	= { 0x00, 0x10 };
CipherSuite	TLS_DHE_DSS_WITH_3DES_EDE_CBC_SHA	= { 0x00, 0x13 };
CipherSuite	TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA	= { 0x00, 0x16 };
CipherSuite	TLS_DH_DSS_WITH_AES_128_CBC_SHA	= { 0x00, 0x30 };
CipherSuite	TLS_DH_RSA_WITH_AES_128_CBC_SHA	= { 0x00, 0x31 };
CipherSuite	TLS_DHE_DSS_WITH_AES_128_CBC_SHA	= { 0x00, 0x32 };
CipherSuite	TLS_DHE_RSA_WITH_AES_128_CBC_SHA	= { 0x00, 0x33 };
CipherSuite	TLS_DH_DSS_WITH_AES_256_CBC_SHA	= { 0x00, 0x36 };
CipherSuite	TLS_DH_RSA_WITH_AES_256_CBC_SHA	= { 0x00, 0x37 };
CipherSuite	TLS_DHE_DSS_WITH_AES_256_CBC_SHA	= { 0x00, 0x38 };
CipherSuite	TLS_DHE_RSA_WITH_AES_256_CBC_SHA	= { 0x00, 0x39 };
CipherSuite	TLS_DH_DSS_WITH_AES_128_CBC_SHA256	= { 0x00, 0x3E };
CipherSuite	TLS_DH_RSA_WITH_AES_128_CBC_SHA256	= { 0x00, 0x3F };
CipherSuite	TLS_DHE_DSS_WITH_AES_128_CBC_SHA256	= { 0x00, 0x40 };
CipherSuite	TLS_DHE_RSA_WITH_AES_128_CBC_SHA256	= { 0x00, 0x67 };
CipherSuite	TLS_DH_DSS_WITH_AES_256_CBC_SHA256	= { 0x00, 0x68 };
CipherSuite	TLS_DH_RSA_WITH_AES_256_CBC_SHA256	= { 0x00, 0x69 };
CipherSuite	TLS_DHE_DSS_WITH_AES_256_CBC_SHA256	= { 0x00, 0x6A };
CipherSuite	TLS_DHE_RSA_WITH_AES_256_CBC_SHA256	= { 0x00, 0x6B };

The following cipher suites are used for completely anonymous Diffie-Hellman communications in which neither party is authenticated. Note that this mode is vulnerable to man-in-the-middle attacks. Using this mode therefore is of limited use: These cipher suites MUST NOT be used by TLS 1.2 implementations unless the application layer has specifically requested to allow anonymous key exchange. (Anonymous key exchange may sometimes be acceptable, for example, to support opportunistic encryption when no set-up for authentication is in place, or when TLS is used as part of more complex security protocols that have other means to ensure authentication.)

CipherSuite	TLS_DH_anon_WITH_RC4_128_MD5	= { 0x00, 0x18 };
CipherSuite	TLS_DH_anon_WITH_3DES_EDE_CBC_SHA	= { 0x00, 0x1B };
CipherSuite	TLS_DH_anon_WITH_AES_128_CBC_SHA	= { 0x00, 0x34 };
CipherSuite	TLS_DH_anon_WITH_AES_256_CBC_SHA	= { 0x00, 0x3A };
CipherSuite	TLS_DH_anon_WITH_AES_128_CBC_SHA256	= { 0x00, 0x6C };
CipherSuite	TLS_DH_anon_WITH_AES_256_CBC_SHA256	= { 0x00, 0x6D };

Note that using non-anonymous key exchange without actually verifying the key exchange is essentially equivalent to anonymous key exchange, and the same precautions apply. While non-anonymous key exchange will generally involve a higher computational and communicational cost than anonymous key exchange, it may be in the interest of interoperability not to disable non-anonymous key exchange when the application layer is allowing anonymous key exchange.

New cipher suite values have been assigned by IANA as described in Section 12.

Note: The cipher suite values { 0x00, 0x1C } and { 0x00, 0x1D } are reserved to avoid collision with Fortezza-based cipher suites in SSL 3.

A.6. The Security Parameters

These security parameters are determined by the TLS Handshake Protocol and provided as parameters to the TLS record layer in order to initialize a connection state. SecurityParameters includes:

```
enum { null(0), (255) } CompressionMethod;

enum { server, client } ConnectionEnd;

enum { tls_prf_sha256 } PRFAlgorithm;

enum { null, rc4, 3des, aes } BulkCipherAlgorithm;

enum { stream, block, aead } CipherType;

enum { null, hmac_md5, hmac_sha1, hmac_sha256, hmac_sha384,
      hmac_sha512 } MACAlgorithm;

/* Other values may be added to the algorithms specified in
CompressionMethod, PRFAlgorithm, BulkCipherAlgorithm, and
MACAlgorithm. */

struct {
    ConnectionEnd          entity;
    PRFAlgorithm           prf_algorithm;
    BulkCipherAlgorithm    bulk_cipher_algorithm;
    CipherType             cipher_type;
    uint8                 enc_key_length;
    uint8                 block_length;
    uint8                 fixed_iv_length;
    uint8                 record_iv_length;
    MACAlgorithm           mac_algorithm;
    uint8                 mac_length;
    uint8                 mac_key_length;
    CompressionMethod      compression_algorithm;
    opaque                 master_secret[48];
    opaque                 client_random[32];
    opaque                 server_random[32];
} SecurityParameters;
```

A.7. Changes to RFC 4492

RFC 4492 [TLSECC] adds Elliptic Curve cipher suites to TLS. This document changes some of the structures used in that document. This section details the required changes for implementors of both RFC 4492 and TLS 1.2. Implementors of TLS 1.2 who are not implementing RFC 4492 do not need to read this section.

This document adds a "signature_algorithm" field to the digitally-signed element in order to identify the signature and digest algorithms used to create a signature. This change applies to digital signatures formed using ECDSA as well, thus allowing ECDSA signatures to be used with digest algorithms other than SHA-1, provided such use is compatible with the certificate and any restrictions imposed by future revisions of [PKIX].

As described in Sections 7.4.2 and 7.4.6, the restrictions on the signature algorithms used to sign certificates are no longer tied to the cipher suite (when used by the server) or the ClientCertificateType (when used by the client). Thus, the restrictions on the algorithm used to sign certificates specified in Sections 2 and 3 of RFC 4492 are also relaxed. As in this document, the restrictions on the keys in the end-entity certificate remain.

Appendix B. Glossary

Advanced Encryption Standard (AES)

AES [AES] is a widely used symmetric encryption algorithm. AES is a block cipher with a 128-, 192-, or 256-bit keys and a 16-byte block size. TLS currently only supports the 128- and 256-bit key sizes.

application protocol

An application protocol is a protocol that normally layers directly on top of the transport layer (e.g., TCP/IP). Examples include HTTP, TELNET, FTP, and SMTP.

asymmetric cipher

See public key cryptography.

authenticated encryption with additional data (AEAD)

A symmetric encryption algorithm that simultaneously provides confidentiality and message integrity.

authentication

Authentication is the ability of one entity to determine the identity of another entity.

block cipher

A block cipher is an algorithm that operates on plaintext in groups of bits, called blocks. 64 bits was, and 128 bits is, a common block size.

bulk cipher

A symmetric encryption algorithm used to encrypt large quantities of data.

cipher block chaining (CBC)

CBC is a mode in which every plaintext block encrypted with a block cipher is first exclusive-ORed with the previous ciphertext block (or, in the case of the first block, with the initialization vector). For decryption, every block is first decrypted, then exclusive-ORed with the previous ciphertext block (or IV).

certificate

As part of the X.509 protocol (a.k.a. ISO Authentication framework), certificates are assigned by a trusted Certificate Authority and provide a strong binding between a party's identity or some other attributes and its public key.

client

The application entity that initiates a TLS connection to a server. This may or may not imply that the client initiated the underlying transport connection. The primary operational difference between the server and client is that the server is generally authenticated, while the client is only optionally authenticated.

client write key

The key used to encrypt data written by the client.

client write MAC key

The secret data used to authenticate data written by the client.

connection

A connection is a transport (in the OSI layering model definition) that provides a suitable type of service. For TLS, such connections are peer-to-peer relationships. The connections are transient. Every connection is associated with one session.

Data Encryption Standard

DES [DES] still is a very widely used symmetric encryption algorithm although it is considered as rather weak now. DES is a block cipher with a 56-bit key and an 8-byte block size. Note that in TLS, for key generation purposes, DES is treated as having an 8-byte key length (64 bits), but it still only provides 56 bits

of protection. (The low bit of each key byte is presumed to be set to produce odd parity in that key byte.) DES can also be operated in a mode [3DES] where three independent keys and three encryptions are used for each block of data; this uses 168 bits of key (24 bytes in the TLS key generation method) and provides the equivalent of 112 bits of security.

Digital Signature Standard (DSS)

A standard for digital signing, including the Digital Signing Algorithm, approved by the National Institute of Standards and Technology, defined in NIST FIPS PUB 186-2, "Digital Signature Standard", published January 2000 by the U.S. Department of Commerce [DSS]. A significant update [DSS-3] has been drafted and was published in March 2006.

digital signatures

Digital signatures utilize public key cryptography and one-way hash functions to produce a signature of the data that can be authenticated, and is difficult to forge or repudiate.

handshake An initial negotiation between client and server that establishes the parameters of their transactions.

Initialization Vector (IV)

When a block cipher is used in CBC mode, the initialization vector is exclusive-ORed with the first plaintext block prior to encryption.

Message Authentication Code (MAC)

A Message Authentication Code is a one-way hash computed from a message and some secret data. It is difficult to forge without knowing the secret data. Its purpose is to detect if the message has been altered.

master secret

Secure secret data used for generating encryption keys, MAC secrets, and IVs.

MD5

MD5 [MD5] is a hashing function that converts an arbitrarily long data stream into a hash of fixed size (16 bytes). Due to significant progress in cryptanalysis, at the time of publication of this document, MD5 no longer can be considered a 'secure' hashing function.

public key cryptography

A class of cryptographic techniques employing two-key ciphers. Messages encrypted with the public key can only be decrypted with the associated private key. Conversely, messages signed with the private key can be verified with the public key.

one-way hash function

A one-way transformation that converts an arbitrary amount of data into a fixed-length hash. It is computationally hard to reverse the transformation or to find collisions. MD5 and SHA are examples of one-way hash functions.

RC4

A stream cipher invented by Ron Rivest. A compatible cipher is described in [SCH].

RSA

A very widely used public key algorithm that can be used for either encryption or digital signing. [RSA]

server

The server is the application entity that responds to requests for connections from clients. See also "client".

session

A TLS session is an association between a client and a server. Sessions are created by the handshake protocol. Sessions define a set of cryptographic security parameters that can be shared among multiple connections. Sessions are used to avoid the expensive negotiation of new security parameters for each connection.

session identifier

A session identifier is a value generated by a server that identifies a particular session.

server write key

The key used to encrypt data written by the server.

server write MAC key

The secret data used to authenticate data written by the server.

SHA

The Secure Hash Algorithm [SHS] is defined in FIPS PUB 180-2. It produces a 20-byte output. Note that all references to SHA (without a numerical suffix) actually use the modified SHA-1 algorithm.

SHA-256

The 256-bit Secure Hash Algorithm is defined in FIPS PUB 180-2. It produces a 32-byte output.

SSL

Netscape's Secure Socket Layer protocol [SSL3]. TLS is based on SSL Version 3.0.

stream cipher

An encryption algorithm that converts a key into a cryptographically strong keystream, which is then exclusive-ORed with the plaintext.

symmetric cipher

See bulk cipher.

Transport Layer Security (TLS)

This protocol; also, the Transport Layer Security working group of the Internet Engineering Task Force (IETF). See "Working Group Information" at the end of this document (see page 99).

Appendix C. Cipher Suite Definitions

Cipher Suite	Key Exchange	Cipher	Mac
TLS_NULL_WITH_NULL_NULL	NULL	NULL	NULL
TLS_RSA_WITH_NULL_MD5	RSA	NULL	MD5
TLS_RSA_WITH_NULL_SHA	RSA	NULL	SHA
TLS_RSA_WITH_NULL_SHA256	RSA	NULL	SHA256
TLS_RSA_WITH_RC4_128_MD5	RSA	RC4_128	MD5
TLS_RSA_WITH_RC4_128_SHA	RSA	RC4_128	SHA
TLS_RSA_WITH_3DES_EDE_CBC_SHA	RSA	3DES_EDE_CBC	SHA
TLS_RSA_WITH_AES_128_CBC_SHA	RSA	AES_128_CBC	SHA
TLS_RSA_WITH_AES_256_CBC_SHA	RSA	AES_256_CBC	SHA
TLS_RSA_WITH_AES_128_CBC_SHA256	RSA	AES_128_CBC	SHA256
TLS_RSA_WITH_AES_256_CBC_SHA256	RSA	AES_256_CBC	SHA256
TLS_DH_DSS_WITH_3DES_EDE_CBC_SHA	DH_DSS	3DES_EDE_CBC	SHA
TLS_DH_RSA_WITH_3DES_EDE_CBC_SHA	DH_RSA	3DES_EDE_CBC	SHA
TLS_DHE_DSS_WITH_3DES_EDE_CBC_SHA	DHE_DSS	3DES_EDE_CBC	SHA
TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA	DHE_RSA	3DES_EDE_CBC	SHA
TLS_DH_anon_WITH_RC4_128_MD5	DH_anon	RC4_128	MD5
TLS_DH_anon_WITH_3DES_EDE_CBC_SHA	DH_anon	3DES_EDE_CBC	SHA
TLS_DH_DSS_WITH_AES_128_CBC_SHA	DH_DSS	AES_128_CBC	SHA
TLS_DH_RSA_WITH_AES_128_CBC_SHA	DH_RSA	AES_128_CBC	SHA
TLS_DHE_DSS_WITH_AES_128_CBC_SHA	DHE_DSS	AES_128_CBC	SHA
TLS_DHE_RSA_WITH_AES_128_CBC_SHA	DHE_RSA	AES_128_CBC	SHA
TLS_DH_anon_WITH_AES_128_CBC_SHA	DH_anon	AES_128_CBC	SHA
TLS_DH_DSS_WITH_AES_256_CBC_SHA	DH_DSS	AES_256_CBC	SHA
TLS_DH_RSA_WITH_AES_256_CBC_SHA	DH_RSA	AES_256_CBC	SHA
TLS_DHE_DSS_WITH_AES_256_CBC_SHA	DHE_DSS	AES_256_CBC	SHA
TLS_DHE_RSA_WITH_AES_256_CBC_SHA	DHE_RSA	AES_256_CBC	SHA
TLS_DH_anon_WITH_AES_256_CBC_SHA	DH_anon	AES_256_CBC	SHA
TLS_DH_DSS_WITH_AES_128_CBC_SHA256	DH_DSS	AES_128_CBC	SHA256
TLS_DH_RSA_WITH_AES_128_CBC_SHA256	DH_RSA	AES_128_CBC	SHA256
TLS_DHE_DSS_WITH_AES_128_CBC_SHA256	DHE_DSS	AES_128_CBC	SHA256
TLS_DHE_RSA_WITH_AES_128_CBC_SHA256	DHE_RSA	AES_128_CBC	SHA256
TLS_DH_anon_WITH_AES_128_CBC_SHA256	DH_anon	AES_128_CBC	SHA256
TLS_DH_DSS_WITH_AES_256_CBC_SHA256	DH_DSS	AES_256_CBC	SHA256
TLS_DH_RSA_WITH_AES_256_CBC_SHA256	DH_RSA	AES_256_CBC	SHA256
TLS_DHE_DSS_WITH_AES_256_CBC_SHA256	DHE_DSS	AES_256_CBC	SHA256
TLS_DHE_RSA_WITH_AES_256_CBC_SHA256	DHE_RSA	AES_256_CBC	SHA256
TLS_DH_anon_WITH_AES_256_CBC_SHA256	DH_anon	AES_256_CBC	SHA256

Cipher	Type	Key Material	IV Size	Block Size
-----	-----	-----	-----	-----
NULL	Stream	0	0	N/A
RC4_128	Stream	16	0	N/A
3DES_EDE_CBC	Block	24	8	8
AES_128_CBC	Block	16	16	16
AES_256_CBC	Block	32	16	16

MAC	Algorithm	mac_length	mac_key_length
-----	-----	-----	-----
NULL	N/A	0	0
MD5	HMAC-MD5	16	16
SHA	HMAC-SHA1	20	20
SHA256	HMAC-SHA256	32	32

Type

Indicates whether this is a stream cipher or a block cipher running in CBC mode.

Key Material

The number of bytes from the key_block that are used for generating the write keys.

IV Size

The amount of data needed to be generated for the initialization vector. Zero for stream ciphers; equal to the block size for block ciphers (this is equal to SecurityParameters.record_iv_length).

Block Size

The amount of data a block cipher enciphers in one chunk; a block cipher running in CBC mode can only encrypt an even multiple of its block size.

Appendix D. Implementation Notes

The TLS protocol cannot prevent many common security mistakes. This section provides several recommendations to assist implementors.

D.1. Random Number Generation and Seeding

TLS requires a cryptographically secure pseudorandom number generator (PRNG). Care must be taken in designing and seeding PRNGs. PRNGs based on secure hash operations, most notably SHA-1, are acceptable, but cannot provide more security than the size of the random number generator state.

To estimate the amount of seed material being produced, add the number of bits of unpredictable information in each seed byte. For example, keystroke timing values taken from a PC compatible's 18.2 Hz timer provide 1 or 2 secure bits each, even though the total size of the counter value is 16 bits or more. Seeding a 128-bit PRNG would thus require approximately 100 such timer values.

[RANDOM] provides guidance on the generation of random values.

D.2. Certificates and Authentication

Implementations are responsible for verifying the integrity of certificates and should generally support certificate revocation messages. Certificates should always be verified to ensure proper signing by a trusted Certificate Authority (CA). The selection and addition of trusted CAs should be done very carefully. Users should be able to view information about the certificate and root CA.

D.3. Cipher Suites

TLS supports a range of key sizes and security levels, including some that provide no or minimal security. A proper implementation will probably not support many cipher suites. For instance, anonymous Diffie-Hellman is strongly discouraged because it cannot prevent man-in-the-middle attacks. Applications should also enforce minimum and maximum key sizes. For example, certificate chains containing 512-bit RSA keys or signatures are not appropriate for high-security applications.

D.4. Implementation Pitfalls

Implementation experience has shown that certain parts of earlier TLS specifications are not easy to understand, and have been a source of interoperability and security problems. Many of these areas have

been clarified in this document, but this appendix contains a short list of the most important things that require special attention from implementors.

TLS protocol issues:

- Do you correctly handle handshake messages that are fragmented to multiple TLS records (see Section 6.2.1)? Including corner cases like a ClientHello that is split to several small fragments? Do you fragment handshake messages that exceed the maximum fragment size? In particular, the certificate and certificate request handshake messages can be large enough to require fragmentation.
- Do you ignore the TLS record layer version number in all TLS records before ServerHello (see Appendix E.1)?
- Do you handle TLS extensions in ClientHello correctly, including omitting the extensions field completely?
- Do you support renegotiation, both client and server initiated? While renegotiation is an optional feature, supporting it is highly recommended.
- When the server has requested a client certificate, but no suitable certificate is available, do you correctly send an empty Certificate message, instead of omitting the whole message (see Section 7.4.6)?

Cryptographic details:

- In the RSA-encrypted Premaster Secret, do you correctly send and verify the version number? When an error is encountered, do you continue the handshake to avoid the Bleichenbacher attack (see Section 7.4.7.1)?
- What countermeasures do you use to prevent timing attacks against RSA decryption and signing operations (see Section 7.4.7.1)?
- When verifying RSA signatures, do you accept both NULL and missing parameters (see Section 4.7)? Do you verify that the RSA padding doesn't have additional data after the hash value? [FI06]
- When using Diffie-Hellman key exchange, do you correctly strip leading zero bytes from the negotiated key (see Section 8.1.2)?
- Does your TLS client check that the Diffie-Hellman parameters sent by the server are acceptable (see Section F.1.1.3)?

- How do you generate unpredictable IVs for CBC mode ciphers (see Section 6.2.3.2)?
- Do you accept long CBC mode padding (up to 255 bytes; see Section 6.2.3.2)?
- How do you address CBC mode timing attacks (Section 6.2.3.2)?
- Do you use a strong and, most importantly, properly seeded random number generator (see Appendix D.1) for generating the premaster secret (for RSA key exchange), Diffie-Hellman private values, the DSA "k" parameter, and other security-critical values?

Appendix E. Backward Compatibility

E.1. Compatibility with TLS 1.0/1.1 and SSL 3.0

Since there are various versions of TLS (1.0, 1.1, 1.2, and any future versions) and SSL (2.0 and 3.0), means are needed to negotiate the specific protocol version to use. The TLS protocol provides a built-in mechanism for version negotiation so as not to bother other protocol components with the complexities of version selection.

TLS versions 1.0, 1.1, and 1.2, and SSL 3.0 are very similar, and use compatible ClientHello messages; thus, supporting all of them is relatively easy. Similarly, servers can easily handle clients trying to use future versions of TLS as long as the ClientHello format remains compatible, and the client supports the highest protocol version available in the server.

A TLS 1.2 client who wishes to negotiate with such older servers will send a normal TLS 1.2 ClientHello, containing { 3, 3 } (TLS 1.2) in ClientHello.client_version. If the server does not support this version, it will respond with a ServerHello containing an older version number. If the client agrees to use this version, the negotiation will proceed as appropriate for the negotiated protocol.

If the version chosen by the server is not supported by the client (or not acceptable), the client MUST send a "protocol_version" alert message and close the connection.

If a TLS server receives a ClientHello containing a version number greater than the highest version supported by the server, it MUST reply according to the highest version supported by the server.

A TLS server can also receive a ClientHello containing a version number smaller than the highest supported version. If the server wishes to negotiate with old clients, it will proceed as appropriate

for the highest version supported by the server that is not greater than `ClientHello.client_version`. For example, if the server supports TLS 1.0, 1.1, and 1.2, and `client_version` is TLS 1.0, the server will proceed with a TLS 1.0 `ServerHello`. If server supports (or is willing to use) only versions greater than `client_version`, it **MUST** send a "protocol_version" alert message and close the connection.

Whenever a client already knows the highest protocol version known to a server (for example, when resuming a session), it **SHOULD** initiate the connection in that native protocol.

Note: some server implementations are known to implement version negotiation incorrectly. For example, there are buggy TLS 1.0 servers that simply close the connection when the client offers a version newer than TLS 1.0. Also, it is known that some servers will refuse the connection if any TLS extensions are included in `ClientHello`. Interoperability with such buggy servers is a complex topic beyond the scope of this document, and may require multiple connection attempts by the client.

Earlier versions of the TLS specification were not fully clear on what the record layer version number (`TLSPlaintext.version`) should contain when sending `ClientHello` (i.e., before it is known which version of the protocol will be employed). Thus, TLS servers compliant with this specification **MUST** accept any value `{03,XX}` as the record layer version number for `ClientHello`.

TLS clients that wish to negotiate with older servers **MAY** send any value `{03,XX}` as the record layer version number. Typical values would be `{03,00}`, the lowest version number supported by the client, and the value of `ClientHello.client_version`. No single value will guarantee interoperability with all old servers, but this is a complex topic beyond the scope of this document.

E.2. Compatibility with SSL 2.0

TLS 1.2 clients that wish to support SSL 2.0 servers **MUST** send version 2.0 `CLIENT-HELLO` messages defined in [SSL2]. The message **MUST** contain the same version number as would be used for ordinary `ClientHello`, and **MUST** encode the supported TLS cipher suites in the `CIPHER-SPECS-DATA` field as described below.

Warning: The ability to send version 2.0 `CLIENT-HELLO` messages will be phased out with all due haste, since the newer `ClientHello` format provides better mechanisms for moving to newer versions and negotiating extensions. TLS 1.2 clients **SHOULD NOT** support SSL 2.0.

However, even TLS servers that do not support SSL 2.0 MAY accept version 2.0 CLIENT-HELLO messages. The message is presented below in sufficient detail for TLS server implementors; the true definition is still assumed to be [SSL2].

For negotiation purposes, 2.0 CLIENT-HELLO is interpreted the same way as a ClientHello with a "null" compression method and no extensions. Note that this message MUST be sent directly on the wire, not wrapped as a TLS record. For the purposes of calculating Finished and CertificateVerify, the msg_length field is not considered to be a part of the handshake message.

```
uint8 V2CipherSpec[3];
struct {
    uint16 msg_length;
    uint8 msg_type;
    Version version;
    uint16 cipher_spec_length;
    uint16 session_id_length;
    uint16 challenge_length;
    V2CipherSpec cipher_specs[V2ClientHello.cipher_spec_length];
    opaque session_id[V2ClientHello.session_id_length];
    opaque challenge[V2ClientHello.challenge_length];
} V2ClientHello;
```

msg_length

The highest bit MUST be 1; the remaining bits contain the length of the following data in bytes.

msg_type

This field, in conjunction with the version field, identifies a version 2 ClientHello message. The value MUST be 1.

version

Equal to ClientHello.client_version.

cipher_spec_length

This field is the total length of the field cipher_specs. It cannot be zero and MUST be a multiple of the V2CipherSpec length (3).

session_id_length

This field MUST have a value of zero for a client that claims to support TLS 1.2.

challenge_length

The length in bytes of the client's challenge to the server to authenticate itself. Historically, permissible values are between 16 and 32 bytes inclusive. When using the SSLv2 backward-compatible handshake the client SHOULD use a 32-byte challenge.

cipher_specs

This is a list of all CipherSpecs the client is willing and able to use. In addition to the 2.0 cipher specs defined in [SSL2], this includes the TLS cipher suites normally sent in ClientHello.cipher_suites, with each cipher suite prefixed by a zero byte. For example, the TLS cipher suite {0x00,0x0A} would be sent as {0x00,0x00,0x0A}.

session_id

This field MUST be empty.

challenge

Corresponds to ClientHello.random. If the challenge length is less than 32, the TLS server will pad the data with leading (note: not trailing) zero bytes to make it 32 bytes long.

Note: Requests to resume a TLS session MUST use a TLS client hello.

E.3. Avoiding Man-in-the-Middle Version Rollback

When TLS clients fall back to Version 2.0 compatibility mode, they MUST use special PKCS#1 block formatting. This is done so that TLS servers will reject Version 2.0 sessions with TLS-capable clients.

When a client negotiates SSL 2.0 but also supports TLS, it MUST set the right-hand (least-significant) 8 random bytes of the PKCS padding (not including the terminal null of the padding) for the RSA encryption of the ENCRYPTED-KEY-DATA field of the CLIENT-MASTER-KEY to 0x03 (the other padding bytes are random).

When a TLS-capable server negotiates SSL 2.0 it SHOULD, after decrypting the ENCRYPTED-KEY-DATA field, check that these 8 padding bytes are 0x03. If they are not, the server SHOULD generate a random value for SECRET-KEY-DATA, and continue the handshake (which will eventually fail since the keys will not match). Note that reporting the error situation to the client could make the server vulnerable to attacks described in [BLEI].

Appendix F. Security Analysis

The TLS protocol is designed to establish a secure connection between a client and a server communicating over an insecure channel. This document makes several traditional assumptions, including that attackers have substantial computational resources and cannot obtain secret information from sources outside the protocol. Attackers are assumed to have the ability to capture, modify, delete, replay, and otherwise tamper with messages sent over the communication channel. This appendix outlines how TLS has been designed to resist a variety of attacks.

F.1. Handshake Protocol

The handshake protocol is responsible for selecting a cipher spec and generating a master secret, which together comprise the primary cryptographic parameters associated with a secure session. The handshake protocol can also optionally authenticate parties who have certificates signed by a trusted certificate authority.

F.1.1. Authentication and Key Exchange

TLS supports three authentication modes: authentication of both parties, server authentication with an unauthenticated client, and total anonymity. Whenever the server is authenticated, the channel is secure against man-in-the-middle attacks, but completely anonymous sessions are inherently vulnerable to such attacks. Anonymous servers cannot authenticate clients. If the server is authenticated, its certificate message must provide a valid certificate chain leading to an acceptable certificate authority. Similarly, authenticated clients must supply an acceptable certificate to the server. Each party is responsible for verifying that the other's certificate is valid and has not expired or been revoked.

The general goal of the key exchange process is to create a `pre_master_secret` known to the communicating parties and not to attackers. The `pre_master_secret` will be used to generate the `master_secret` (see Section 8.1). The `master_secret` is required to generate the Finished messages, encryption keys, and MAC keys (see Sections 7.4.9 and 6.3). By sending a correct Finished message, parties thus prove that they know the correct `pre_master_secret`.

F.1.1.1. Anonymous Key Exchange

Completely anonymous sessions can be established using Diffie-Hellman for key exchange. The server's public parameters are contained in the server key exchange message, and the client's are sent in the

client key exchange message. Eavesdroppers who do not know the private values should not be able to find the Diffie-Hellman result (i.e., the `pre_master_secret`).

Warning: Completely anonymous connections only provide protection against passive eavesdropping. Unless an independent tamper-proof channel is used to verify that the Finished messages were not replaced by an attacker, server authentication is required in environments where active man-in-the-middle attacks are a concern.

F.1.1.2. RSA Key Exchange and Authentication

With RSA, key exchange and server authentication are combined. The public key is contained in the server's certificate. Note that compromise of the server's static RSA key results in a loss of confidentiality for all sessions protected under that static key. TLS users desiring Perfect Forward Secrecy should use DHE cipher suites. The damage done by exposure of a private key can be limited by changing one's private key (and certificate) frequently.

After verifying the server's certificate, the client encrypts a `pre_master_secret` with the server's public key. By successfully decoding the `pre_master_secret` and producing a correct Finished message, the server demonstrates that it knows the private key corresponding to the server certificate.

When RSA is used for key exchange, clients are authenticated using the certificate verify message (see Section 7.4.8). The client signs a value derived from all preceding handshake messages. These handshake messages include the server certificate, which binds the signature to the server, and `ServerHello.random`, which binds the signature to the current handshake process.

F.1.1.3. Diffie-Hellman Key Exchange with Authentication

When Diffie-Hellman key exchange is used, the server can either supply a certificate containing fixed Diffie-Hellman parameters or use the server key exchange message to send a set of temporary Diffie-Hellman parameters signed with a DSA or RSA certificate. Temporary parameters are hashed with the `hello.random` values before signing to ensure that attackers do not replay old parameters. In either case, the client can verify the certificate or signature to ensure that the parameters belong to the server.

If the client has a certificate containing fixed Diffie-Hellman parameters, its certificate contains the information required to complete the key exchange. Note that in this case the client and server will generate the same Diffie-Hellman result (i.e.,

pre_master_secret) every time they communicate. To prevent the pre_master_secret from staying in memory any longer than necessary, it should be converted into the master_secret as soon as possible. Client Diffie-Hellman parameters must be compatible with those supplied by the server for the key exchange to work.

If the client has a standard DSA or RSA certificate or is unauthenticated, it sends a set of temporary parameters to the server in the client key exchange message, then optionally uses a certificate verify message to authenticate itself.

If the same DH keypair is to be used for multiple handshakes, either because the client or server has a certificate containing a fixed DH keypair or because the server is reusing DH keys, care must be taken to prevent small subgroup attacks. Implementations SHOULD follow the guidelines found in [SUBGROUP].

Small subgroup attacks are most easily avoided by using one of the DHE cipher suites and generating a fresh DH private key (X) for each handshake. If a suitable base (such as 2) is chosen, $g^X \bmod p$ can be computed very quickly; therefore, the performance cost is minimized. Additionally, using a fresh key for each handshake provides Perfect Forward Secrecy. Implementations SHOULD generate a new X for each handshake when using DHE cipher suites.

Because TLS allows the server to provide arbitrary DH groups, the client should verify that the DH group is of suitable size as defined by local policy. The client SHOULD also verify that the DH public exponent appears to be of adequate size. [KEYSIZ] provides a useful guide to the strength of various group sizes. The server MAY choose to assist the client by providing a known group, such as those defined in [IKEALG] or [MODP]. These can be verified by simple comparison.

F.1.2. Version Rollback Attacks

Because TLS includes substantial improvements over SSL Version 2.0, attackers may try to make TLS-capable clients and servers fall back to Version 2.0. This attack can occur if (and only if) two TLS-capable parties use an SSL 2.0 handshake.

Although the solution using non-random PKCS #1 block type 2 message padding is inelegant, it provides a reasonably secure way for Version 3.0 servers to detect the attack. This solution is not secure against attackers who can brute-force the key and substitute a new ENCRYPTED-KEY-DATA message containing the same key (but with normal padding) before the application-specified wait threshold has expired. Altering the padding of the least-significant 8 bytes of the PKCS

padding does not impact security for the size of the signed hashes and RSA key lengths used in the protocol, since this is essentially equivalent to increasing the input block size by 8 bytes.

F.1.3. Detecting Attacks Against the Handshake Protocol

An attacker might try to influence the handshake exchange to make the parties select different encryption algorithms than they would normally choose.

For this attack, an attacker must actively change one or more handshake messages. If this occurs, the client and server will compute different values for the handshake message hashes. As a result, the parties will not accept each others' Finished messages. Without the master_secret, the attacker cannot repair the Finished messages, so the attack will be discovered.

F.1.4. Resuming Sessions

When a connection is established by resuming a session, new ClientHello.random and ServerHello.random values are hashed with the session's master_secret. Provided that the master_secret has not been compromised and that the secure hash operations used to produce the encryption keys and MAC keys are secure, the connection should be secure and effectively independent from previous connections. Attackers cannot use known encryption keys or MAC secrets to compromise the master_secret without breaking the secure hash operations.

Sessions cannot be resumed unless both the client and server agree. If either party suspects that the session may have been compromised, or that certificates may have expired or been revoked, it should force a full handshake. An upper limit of 24 hours is suggested for session ID lifetimes, since an attacker who obtains a master_secret may be able to impersonate the compromised party until the corresponding session ID is retired. Applications that may be run in relatively insecure environments should not write session IDs to stable storage.

F.2. Protecting Application Data

The master_secret is hashed with the ClientHello.random and ServerHello.random to produce unique data encryption keys and MAC secrets for each connection.

Outgoing data is protected with a MAC before transmission. To prevent message replay or modification attacks, the MAC is computed from the MAC key, the sequence number, the message length, the

message contents, and two fixed character strings. The message type field is necessary to ensure that messages intended for one TLS record layer client are not redirected to another. The sequence number ensures that attempts to delete or reorder messages will be detected. Since sequence numbers are 64 bits long, they should never overflow. Messages from one party cannot be inserted into the other's output, since they use independent MAC keys. Similarly, the server write and client write keys are independent, so stream cipher keys are used only once.

If an attacker does break an encryption key, all messages encrypted with it can be read. Similarly, compromise of a MAC key can make message-modification attacks possible. Because MACs are also encrypted, message-alteration attacks generally require breaking the encryption algorithm as well as the MAC.

Note: MAC keys may be larger than encryption keys, so messages can remain tamper resistant even if encryption keys are broken.

F.3. Explicit IVs

[CBCATT] describes a chosen plaintext attack on TLS that depends on knowing the IV for a record. Previous versions of TLS [TLS1.0] used the CBC residue of the previous record as the IV and therefore enabled this attack. This version uses an explicit IV in order to protect against this attack.

F.4. Security of Composite Cipher Modes

TLS secures transmitted application data via the use of symmetric encryption and authentication functions defined in the negotiated cipher suite. The objective is to protect both the integrity and confidentiality of the transmitted data from malicious actions by active attackers in the network. It turns out that the order in which encryption and authentication functions are applied to the data plays an important role for achieving this goal [ENCAUTH].

The most robust method, called encrypt-then-authenticate, first applies encryption to the data and then applies a MAC to the ciphertext. This method ensures that the integrity and confidentiality goals are obtained with ANY pair of encryption and MAC functions, provided that the former is secure against chosen plaintext attacks and that the MAC is secure against chosen-message attacks. TLS uses another method, called authenticate-then-encrypt, in which first a MAC is computed on the plaintext and then the concatenation of plaintext and MAC is encrypted. This method has been proven secure for CERTAIN combinations of encryption functions and MAC functions, but it is not guaranteed to be secure in general.

In particular, it has been shown that there exist perfectly secure encryption functions (secure even in the information-theoretic sense) that combined with any secure MAC function, fail to provide the confidentiality goal against an active attack. Therefore, new cipher suites and operation modes adopted into TLS need to be analyzed under the authenticate-then-encrypt method to verify that they achieve the stated integrity and confidentiality goals.

Currently, the security of the authenticate-then-encrypt method has been proven for some important cases. One is the case of stream ciphers in which a computationally unpredictable pad of the length of the message, plus the length of the MAC tag, is produced using a pseudorandom generator and this pad is exclusive-ORed with the concatenation of plaintext and MAC tag. The other is the case of CBC mode using a secure block cipher. In this case, security can be shown if one applies one CBC encryption pass to the concatenation of plaintext and MAC and uses a new, independent, and unpredictable IV for each new pair of plaintext and MAC. In versions of TLS prior to 1.1, CBC mode was used properly EXCEPT that it used a predictable IV in the form of the last block of the previous ciphertext. This made TLS open to chosen plaintext attacks. This version of the protocol is immune to those attacks. For exact details in the encryption modes proven secure, see [ENCAUTH].

F.5. Denial of Service

TLS is susceptible to a number of denial-of-service (DoS) attacks. In particular, an attacker who initiates a large number of TCP connections can cause a server to consume large amounts of CPU for doing RSA decryption. However, because TLS is generally used over TCP, it is difficult for the attacker to hide his point of origin if proper TCP SYN randomization is used [SEQNUM] by the TCP stack.

Because TLS runs over TCP, it is also susceptible to a number of DoS attacks on individual connections. In particular, attackers can forge RSTs, thereby terminating connections, or forge partial TLS records, thereby causing the connection to stall. These attacks cannot in general be defended against by a TCP-using protocol. Implementors or users who are concerned with this class of attack should use IPsec AH [AH] or ESP [ESP].

F.6. Final Notes

For TLS to be able to provide a secure connection, both the client and server systems, keys, and applications must be secure. In addition, the implementation must be free of security errors.

The system is only as strong as the weakest key exchange and authentication algorithm supported, and only trustworthy cryptographic functions should be used. Short public keys and anonymous servers should be used with great caution. Implementations and users must be careful when deciding which certificates and certificate authorities are acceptable; a dishonest certificate authority can do tremendous damage.

Normative References

- [AES] National Institute of Standards and Technology, "Specification for the Advanced Encryption Standard (AES)" FIPS 197. November 26, 2001.
- [3DES] National Institute of Standards and Technology, "Recommendation for the Triple Data Encryption Algorithm (TDEA) Block Cipher", NIST Special Publication 800-67, May 2004.
- [DSS] NIST FIPS PUB 186-2, "Digital Signature Standard", National Institute of Standards and Technology, U.S. Department of Commerce, 2000.
- [HMAC] Krawczyk, H., Bellare, M., and R. Canetti, "HMAC: Keyed-Hashing for Message Authentication", RFC 2104, February 1997.
- [MD5] Rivest, R., "The MD5 Message-Digest Algorithm", RFC 1321, April 1992.
- [PKCS1] Jonsson, J. and B. Kaliski, "Public-Key Cryptography Standards (PKCS) #1: RSA Cryptography Specifications Version 2.1", RFC 3447, February 2003.
- [PKIX] Housley, R., Polk, W., Ford, W., and D. Solo, "Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile", RFC 3280, April 2002.
- [SCH] B. Schneier. "Applied Cryptography: Protocols, Algorithms, and Source Code in C, 2nd ed.", Published by John Wiley & Sons, Inc. 1996.
- [SHS] NIST FIPS PUB 180-2, "Secure Hash Standard", National Institute of Standards and Technology, U.S. Department of Commerce, August 2002.

- [REQ] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997.
- [RFC2434] Narten, T. and H. Alvestrand, "Guidelines for Writing an IANA Considerations Section in RFCs", BCP 26, RFC 2434, October 1998.
- [X680] ITU-T Recommendation X.680 (2002) | ISO/IEC 8824-1:2002, Information technology - Abstract Syntax Notation One (ASN.1): Specification of basic notation.
- [X690] ITU-T Recommendation X.690 (2002) | ISO/IEC 8825-1:2002, Information technology - ASN.1 encoding Rules: Specification of Basic Encoding Rules (BER), Canonical Encoding Rules (CER) and Distinguished Encoding Rules (DER).

Informative References

- [AEAD] McGrew, D., "An Interface and Algorithms for Authenticated Encryption", RFC 5116, January 2008.
- [AH] Kent, S., "IP Authentication Header", RFC 4302, December 2005.
- [BLEI] Bleichenbacher D., "Chosen Ciphertext Attacks against Protocols Based on RSA Encryption Standard PKCS #1" in Advances in Cryptology -- CRYPTO'98, LNCS vol. 1462, pages: 1-12, 1998.
- [CBCATT] Moeller, B., "Security of CBC Ciphersuites in SSL/TLS: Problems and Countermeasures", <http://www.openssl.org/~bodo/tls-cbc.txt>.
- [CBCTIME] Canvel, B., Hiltgen, A., Vaudenay, S., and M. Vuagnoux, "Password Interception in a SSL/TLS Channel", Advances in Cryptology -- CRYPTO 2003, LNCS vol. 2729, 2003.
- [CCM] "NIST Special Publication 800-38C: The CCM Mode for Authentication and Confidentiality", <http://csrc.nist.gov/publications/nistpubs/800-38C/SP800-38C.pdf>
- [DES] National Institute of Standards and Technology, "Data Encryption Standard (DES)", FIPS PUB 46-3, October 1999.

- [DSS-3] NIST FIPS PUB 186-3 Draft, "Digital Signature Standard", National Institute of Standards and Technology, U.S. Department of Commerce, 2006.
- [ECDSA] American National Standards Institute, "Public Key Cryptography for the Financial Services Industry: The Elliptic Curve Digital Signature Algorithm (ECDSA)", ANSI X9.62-2005, November 2005.
- [ENCAUTH] Krawczyk, H., "The Order of Encryption and Authentication for Protecting Communications (Or: How Secure is SSL?)", Crypto 2001.
- [ESP] Kent, S., "IP Encapsulating Security Payload (ESP)", RFC 4303, December 2005.
- [FI06] Hal Finney, "Bleichenbacher's RSA signature forgery based on implementation error", ietf-openpgp@imc.org mailing list, 27 August 2006, <http://www.imc.org/ietf-openpgp/mail-archive/msg14307.html>.
- [GCM] Dworkin, M., NIST Special Publication 800-38D, "Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC", November 2007.
- [IKEALG] Schiller, J., "Cryptographic Algorithms for Use in the Internet Key Exchange Version 2 (IKEv2)", RFC 4307, December 2005.
- [KEYSIZ] Orman, H. and P. Hoffman, "Determining Strengths For Public Keys Used For Exchanging Symmetric Keys", BCP 86, RFC 3766, April 2004.
- [KPR03] Klima, V., Pokorny, O., Rosa, T., "Attacking RSA-based Sessions in SSL/TLS", <http://eprint.iacr.org/2003/052/>, March 2003.
- [MODP] Kivinen, T. and M. Kojo, "More Modular Exponential (MODP) Diffie-Hellman groups for Internet Key Exchange (IKE)", RFC 3526, May 2003.
- [PKCS6] RSA Laboratories, "PKCS #6: RSA Extended Certificate Syntax Standard", version 1.5, November 1993.
- [PKCS7] RSA Laboratories, "PKCS #7: RSA Cryptographic Message Syntax Standard", version 1.5, November 1993.

- [RANDOM] Eastlake, D., 3rd, Schiller, J., and S. Crocker, "Randomness Requirements for Security", BCP 106, RFC 4086, June 2005.
- [RFC3749] Hollenbeck, S., "Transport Layer Security Protocol Compression Methods", RFC 3749, May 2004.
- [RFC4366] Blake-Wilson, S., Nystrom, M., Hopwood, D., Mikkelsen, J., and T. Wright, "Transport Layer Security (TLS) Extensions", RFC 4366, April 2006.
- [RSA] R. Rivest, A. Shamir, and L. M. Adleman, "A Method for Obtaining Digital Signatures and Public-Key Cryptosystems", Communications of the ACM, v. 21, n. 2, Feb 1978, pp. 120-126.
- [SEQNUM] Bellare, S., "Defending Against Sequence Number Attacks", RFC 1948, May 1996.
- [SSL2] Hickman, Kipp, "The SSL Protocol", Netscape Communications Corp., Feb 9, 1995.
- [SSL3] A. Freier, P. Karlton, and P. Kocher, "The SSL 3.0 Protocol", Netscape Communications Corp., Nov 18, 1996.
- [SUBGROUP] Zuccherato, R., "Methods for Avoiding the "Small-Subgroup" Attacks on the Diffie-Hellman Key Agreement Method for S/MIME", RFC 2785, March 2000.
- [TCP] Postel, J., "Transmission Control Protocol", STD 7, RFC 793, September 1981.
- [TIMING] Boneh, D., Brumley, D., "Remote timing attacks are practical", USENIX Security Symposium 2003.
- [TLSAES] Chown, P., "Advanced Encryption Standard (AES) Ciphersuites for Transport Layer Security (TLS)", RFC 3268, June 2002.
- [TLSECC] Blake-Wilson, S., Bolyard, N., Gupta, V., Hawk, C., and B. Moeller, "Elliptic Curve Cryptography (ECC) Cipher Suites for Transport Layer Security (TLS)", RFC 4492, May 2006.
- [TLSEXT] Eastlake, D., 3rd, "Transport Layer Security (TLS) Extensions: Extension Definitions", Work in Progress, February 2008.

- [TLSPGP] Mavrogiannopoulos, N., "Using OpenPGP Keys for Transport Layer Security (TLS) Authentication", RFC 5081, November 2007.
- [TLSPSK] Eronen, P., Ed., and H. Tschofenig, Ed., "Pre-Shared Key Ciphersuites for Transport Layer Security (TLS)", RFC 4279, December 2005.
- [TLS1.0] Dierks, T. and C. Allen, "The TLS Protocol Version 1.0", RFC 2246, January 1999.
- [TLS1.1] Dierks, T. and E. Rescorla, "The Transport Layer Security (TLS) Protocol Version 1.1", RFC 4346, April 2006.
- [X501] ITU-T Recommendation X.501: Information Technology - Open Systems Interconnection - The Directory: Models, 1993.
- [XDR] Eisler, M., Ed., "XDR: External Data Representation Standard", STD 67, RFC 4506, May 2006.

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